CEBU INSTITUTE OF TECHNOLOGY UNIVERSITY

COLLEGE OF COMPUTER STUDIES

Software Design Description

for

IntelliTrack: A Web-Based AI-Powered Capstone Deliverables Submission Tracking and Analytics System

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Signature

Change History

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Module 1 *8*
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1. Introduction

1.1. Purpose

This Software Design Description (SDD) document describes the system design and technical architecture of **IntelliTrack**, a web-based capstone deliverables tracking and analytics platform.

The document defines how the system components interact, how functional requirements from the SRS are implemented, and how data flows through the system. It provides detailed explanations of system modules, backend services, user interfaces, and database design.

The goal of the design is to create a scalable and maintainable platform capable of monitoring capstone project milestones, tracking deliverable timelines, and generating analytics insights that support academic monitoring and decision-making.

1.2. Scope

IntelliTrack is designed as a web-based system that assists academic institutions in monitoring the progress of capstone projects through structured deliverable tracking and analytics.

The system enables students, advisers, coordinators, and administrators to access a centralized platform where capstone deliverables are monitored according to defined milestones and deadlines.

The platform supports the following core capabilities:

- Secure role-based user authentication
- Class and deliverable configuration by administrators
- Monitoring of milestone deadlines and submission timelines

- AI-assisted reminder notifications
- Data analytics dashboards for submission insights
- Institutional reporting for monitoring capstone performance

Through these capabilities, IntelliTrack provides real-time visibility of project progress and helps academic staff monitor compliance with required deliverables

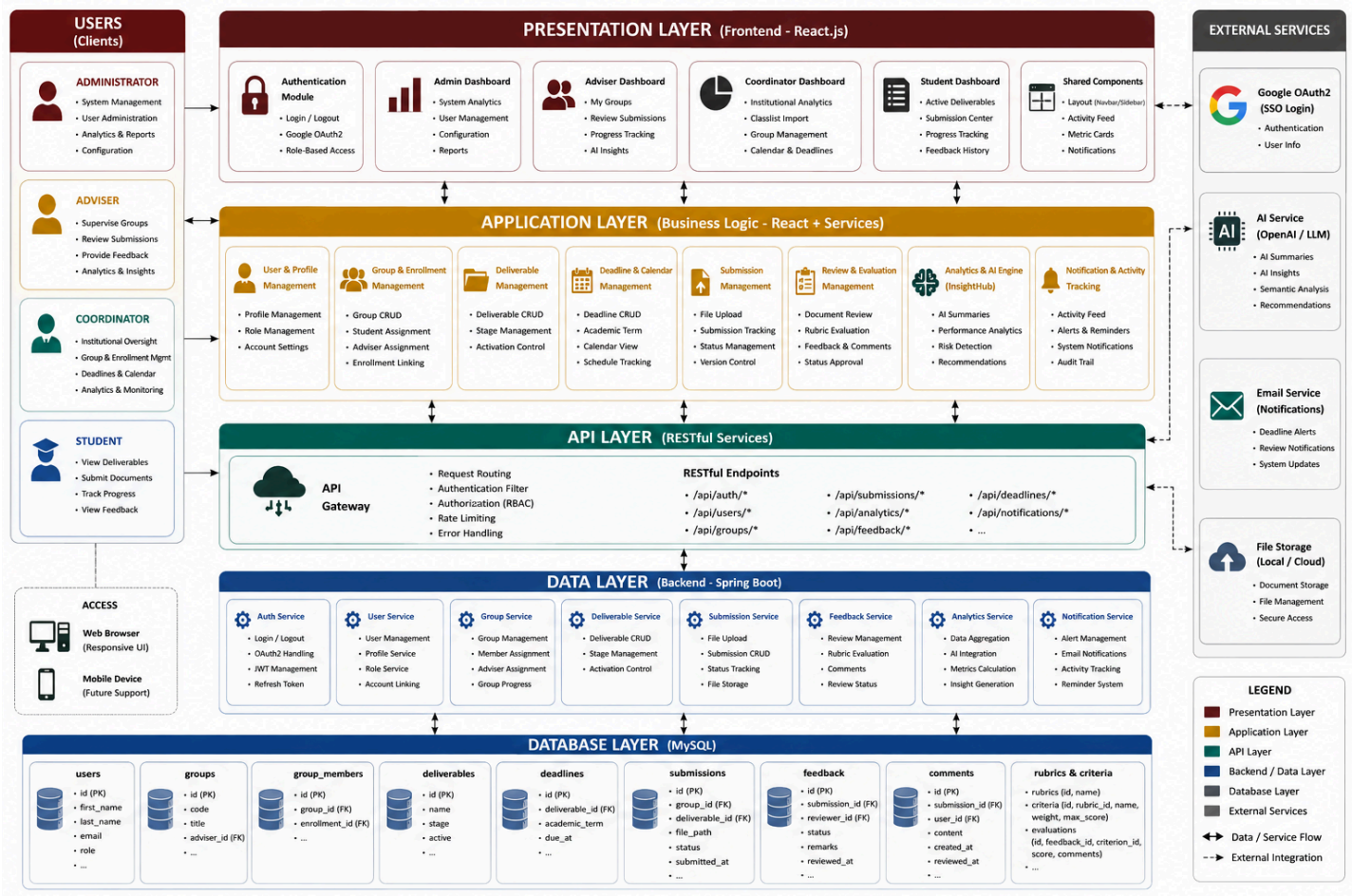
1.3. Definitions and Acronyms

Acronym & Abbreviations	**Definition**
AI	Artificial Intelligence
RBAC	Role-Based Access Control
LMS	Learning Management System
UI	User Interface
UX	User Experience
CRUD	Create, Read, Update, Delete (basic operations in software systems)
CDSS	Capstone Deliverables Submission System
Milestone	A required capstone deliverable with a specific deadline (e.g., Proposal, Midterm, Final)
Version Control	A system that tracks document versions, changes, and submission histories
Analytics Dashboard	A visual interface showing real-time submission status and progress insights

1.4. References

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2. Architectural Design

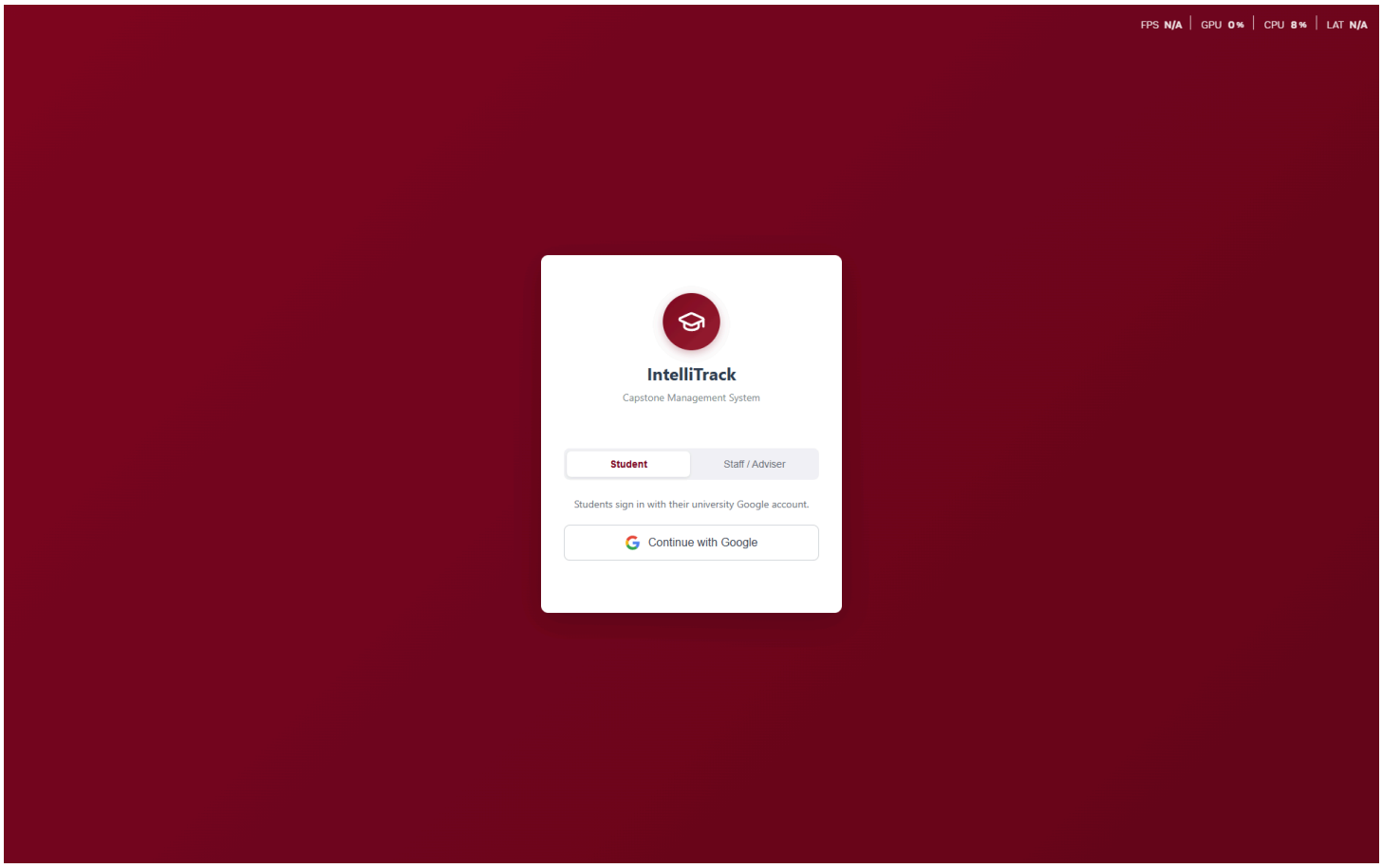


3. Detailed Design

Module 1: User Management System

1.1 Log In

- User Interface Design



- Front-end component(s)
    - **Login**
        - Description and purpose:
            - Provides the user interface where users initiate the login process.
        - Component type or format:
            - React Component (components/Login.jsx)
    - **AuthContext, AuthProvider, and useAuth**

        - Description and purpose:
            - Handles the frontend authentication workflow of the system. It manages user login sessions, token storage, OAuth callback handling, logout functionality, role validation, and authentication state persistence using local storage. It also provides authentication data and utility functions that can be accessed across the entire frontend application.
        - Component type or format:
            - React Authentication Context (context/AuthContext.jsx)

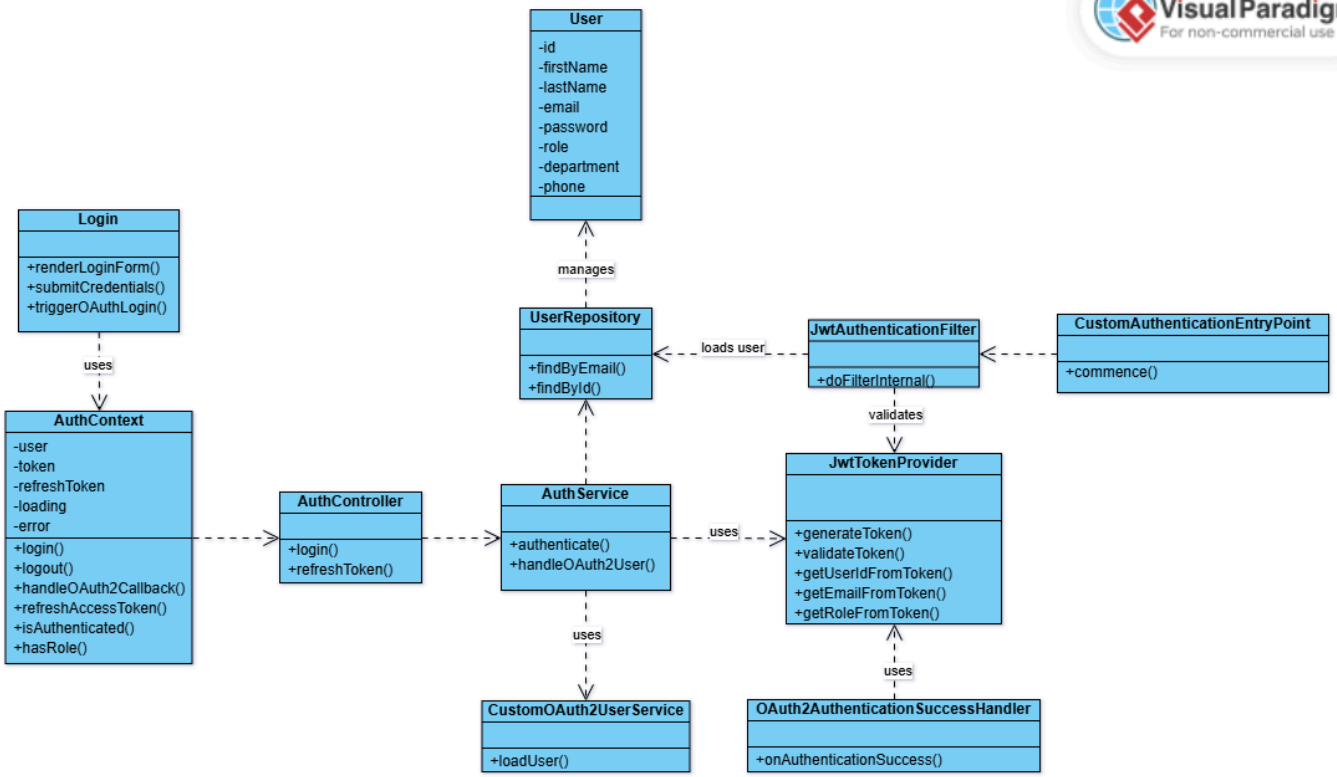
- Back-end component(s)
    - **User**
        - Description and purpose
            - Represents the authenticated system user entity. It stores user account information including personal details, email credentials, assigned roles, and associated project group information.
        - Component type or format
            - Entity Class (entity/User.java)
    - **UserRepository**
        - Description and purpose
            - Retrieves user records from the database based on the authenticated email. It is also responsible for accessing user roles and account details.
        - Component type or format
            - Repository Interface (repository/UserRepository.java)
    - **JwtTokenProvider**
        - Description and purpose
            - Generates and validates JSON Web Tokens (JWT) used for user authentication and session management. It extracts authenticated user information such as user ID, email, and role from generated tokens.
        - Component type or format

- Security Service(security/JwtTokenProvider.java)

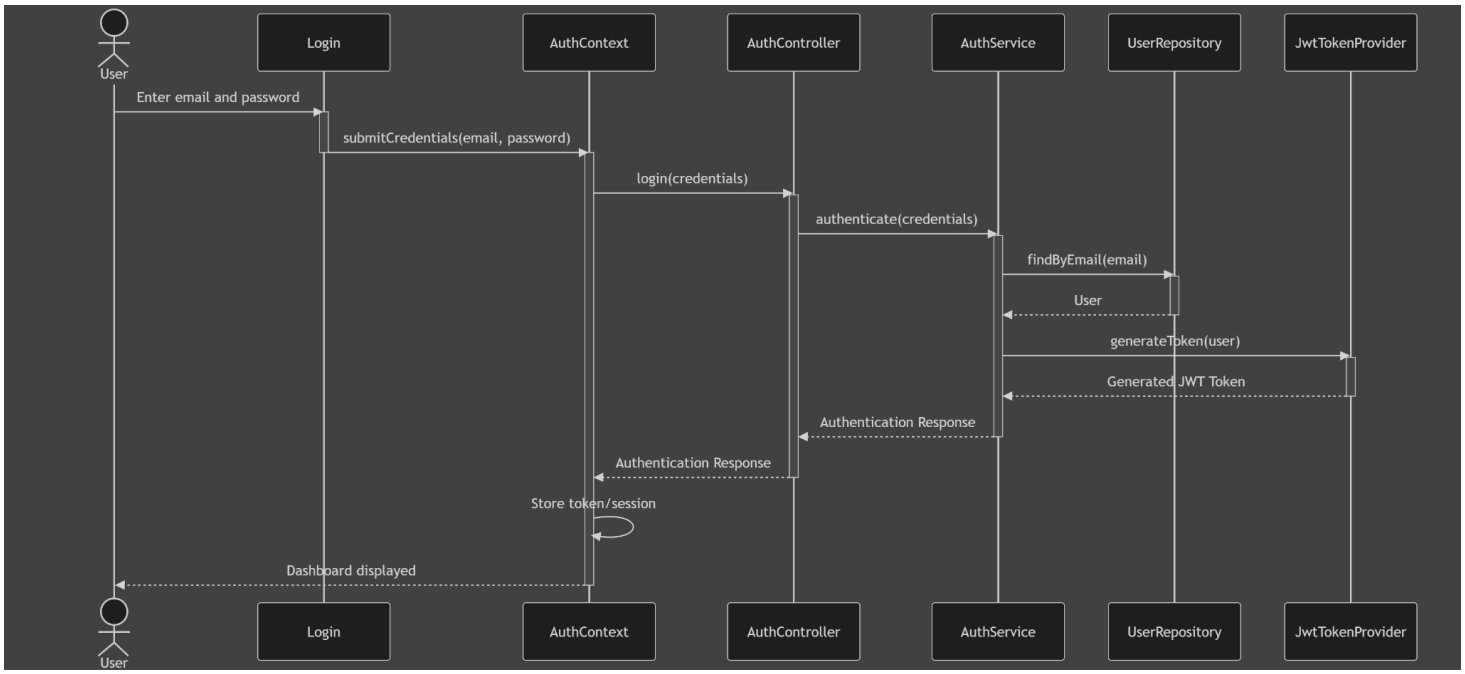
- **JwtAuthenticationFilter**
  - Description and purpose
    - Intercepts incoming HTTP requests and validates JWT authentication tokens before allowing access to protected system resources. It ensures only authenticated users can access secured API endpoints.
  - Component type or format
    - Security Filter (security/JwtAuthenticationFilter.java)
- **CustomAuthenticationEntryPoint**
  - Description and purpose
    - Handles unauthorized access attempts to secure resources. It returns appropriate authentication error responses for API requests and redirects unauthenticated users to Google OAuth login when necessary.
  - Component type or format
    - Security Authentication Entry Point (security/CustomAuthenticationEntryPoint.java)
- **OAuth2AuthenticationSuccessHandler**
  - Description and purpose
    - Processes successful Google OAuth authentication events. It handles authenticated user sessions and redirects authenticated users to the frontend application after successful login.
  - Component type or format
    - OAuth2 Authentication Handler (security/OAuth2AuthenticationSuccessHandler.java)
- **OAuth2AuthenticationFailureHandler**
  - Description and purpose
    - Handles failed Google OAuth authentication attempts by redirecting users to the frontend login page with authentication error indicators. .

- Component type or format
    - OAuth2 Authentication Handler (security/OAuth2AuthenticationFailureHandler.java)

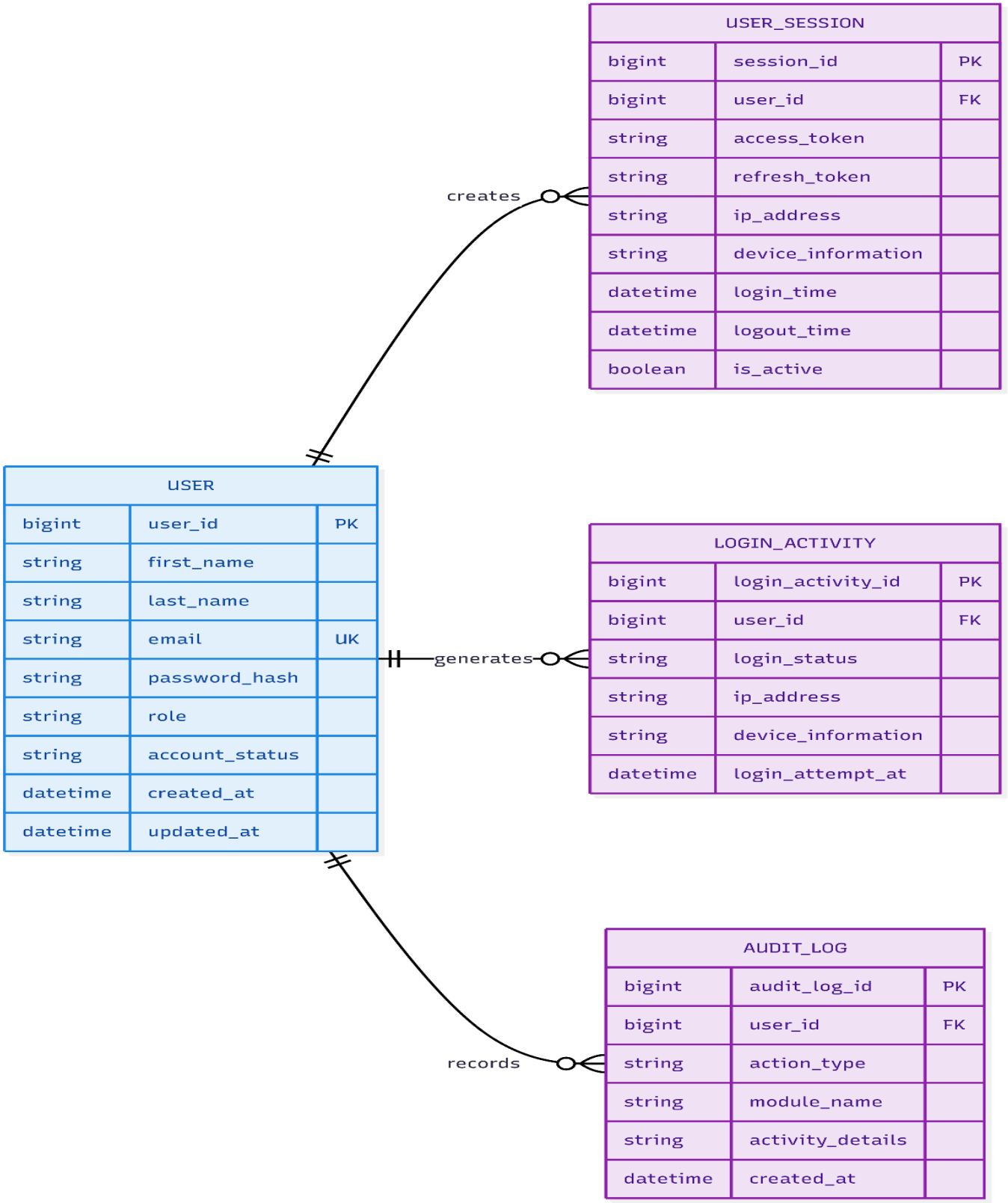
- Object-Oriented Components
    - Class Diagram



    - Sequence Diagram

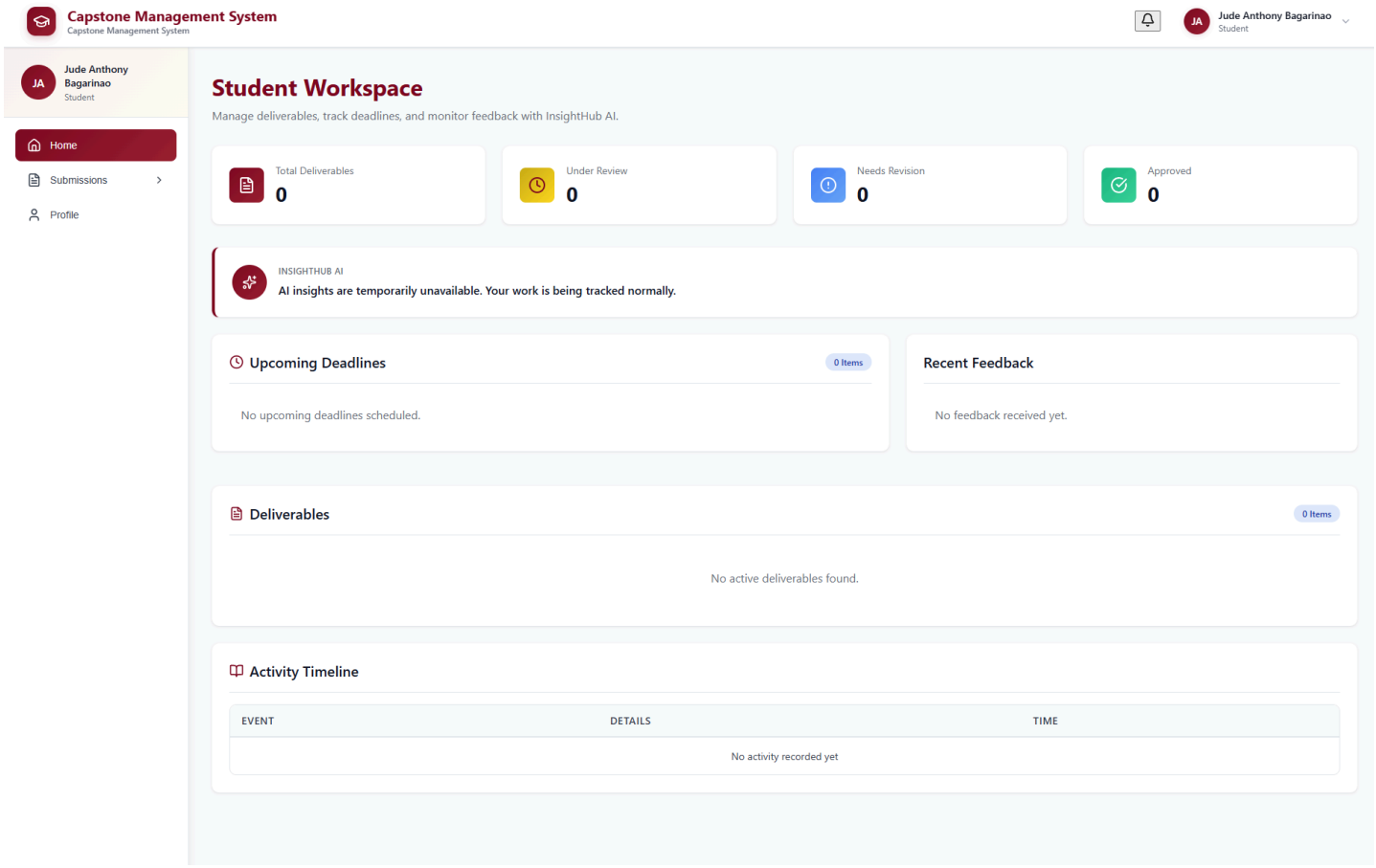


- Data Design
    - ERD or schema

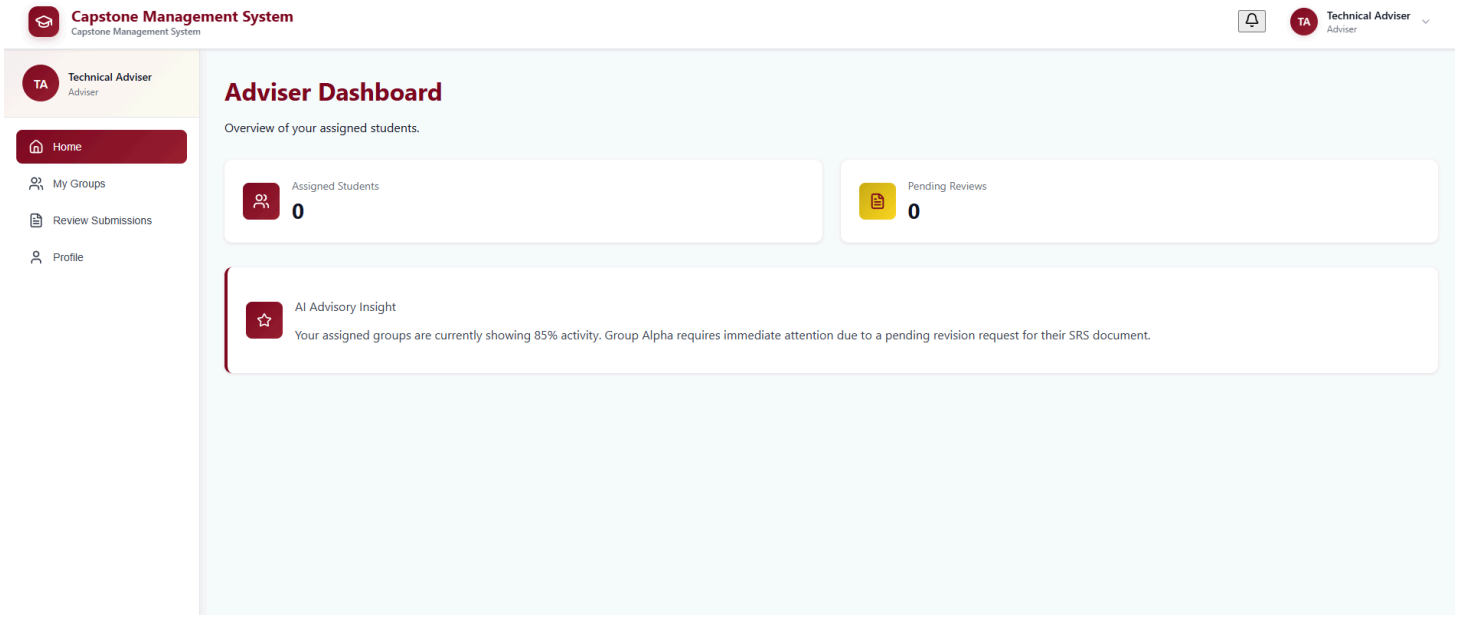

USER_SESSION
bigint session_id PK
bigint user_id FK
string access_token
string refresh_token
string ip_address
string device_information
datetime login_time
datetime logout_time
boolean is_active
creates
USER
bigint user_id PK
string first_name
string last_name
string email UK
string password_hash
string role
string account_status
datetime created_at
datetime updated_at
generates
LOGIN_ACTIVITY
bigint login_activity_id PK
bigint user_id FK
string login_status
string ip_address
string device_information
datetime login_attempt_at
records
AUDIT_LOG
bigint audit_log_id PK
bigint user_id FK
string action_type
string module_name
string activity_details
datetime created_at

1.2 Role Identification

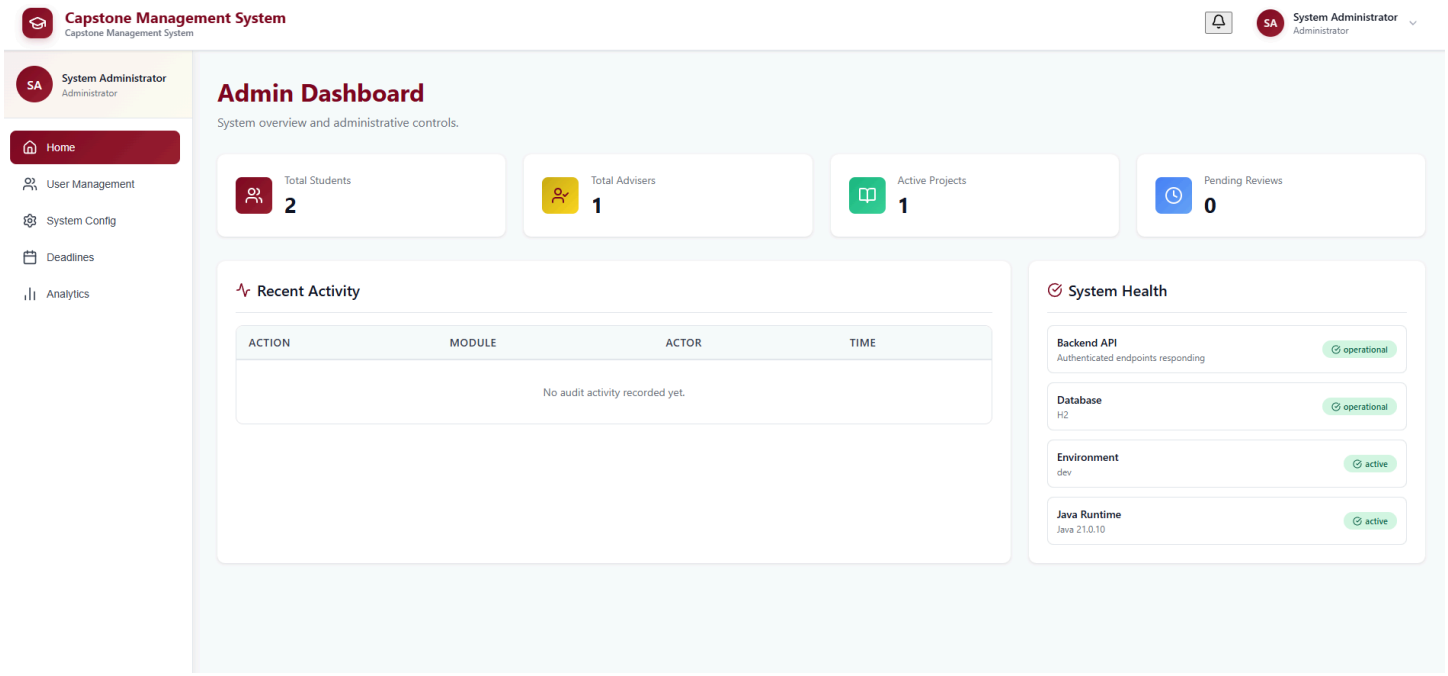
- User Interface Design
    - Student



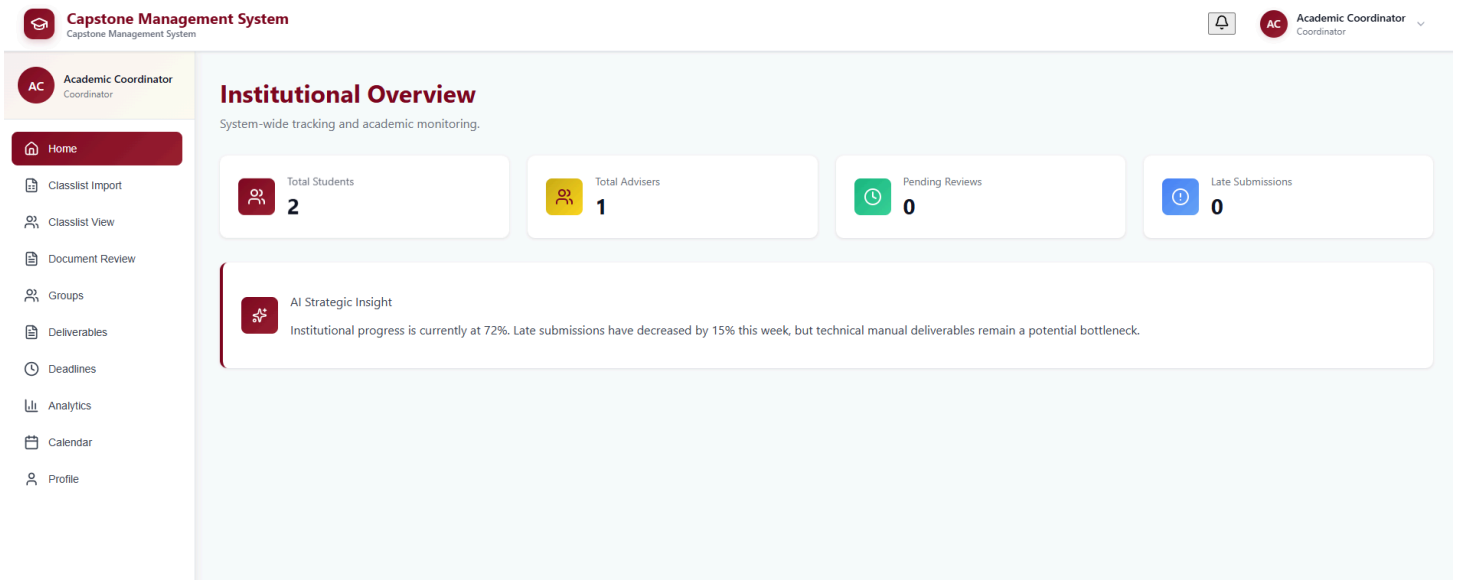
    - Advisor



- Administrator



- Coordinator



- Front-end component(s)
    - **PrivateRoute**
        - Description and purpose
            - Protects system routes that require authenticated access. It verifies whether the user is logged in and checks if the user has the required role before allowing access to protected pages. Unauthorized users are redirected to the appropriate page.
        - Component type or format
            - React Route Protection Component (components/PrivateRoute.jsx)
    - **StudentDashboard**
        - Description and purpose
            - Provides the navigation layout and access interface for student users. It dynamically loads active deliverables and allows students to access submission-related modules and profile management pages.
        - Component type or format
            - React Dashboard Component (pages/StudentDashboard.jsx)
    - **AdviserDashboard**
        - Description and purpose
            - Provides adviser-specific access to protected modules such as group management, submission review, and adviser profile pages.

        - Component type or format
            - React Dashboard Component (pages/AdviserDashboard.jsx)
    - **CoordinatorDashboard**
        - Description and purpose
            - Provides coordinator-specific access to institutional monitoring, deliverable management, deadlines, analytics, and classlist modules.
            - React Dashboard Component (pages/CoordinatorDashboard.jsx)
    - **AdminDashboard**
        - Description and purpose
            - Provides administrator-specific access to protected system modules and routes based on authenticated administrator privileges.
        - Component type or format
            - React Dashboard Component (pages/AdminDashboard.jsx)

- Back-end component(s)
    - **JwtTokenProvider**
        - Description and purpose
            - Generates and validates JWT authentication tokens used for role-based access control. It extracts user role information from authentication tokens for authorization checking.
        - Component type or format
            - Security Service (security/JwtTokenProvider.java)
    - **JwtAuthenticationFilter**
        - Description and purpose
            - Intercepts incoming API requests and validates authentication tokens before protected resources are accessed. It ensures that authenticated users possess valid sessions and authorized roles.
        - Component type or format
            - Security Filter (security/JwtAuthenticationFilter.java)
    - **UserRepository**
        - Description and purpose
            - Retrieves authenticated user records and role information from the database. It supports role-based authorization by accessing stored user account data.

- Component type or format
  - Repository Interface (repository/UserRepository.java)

o User

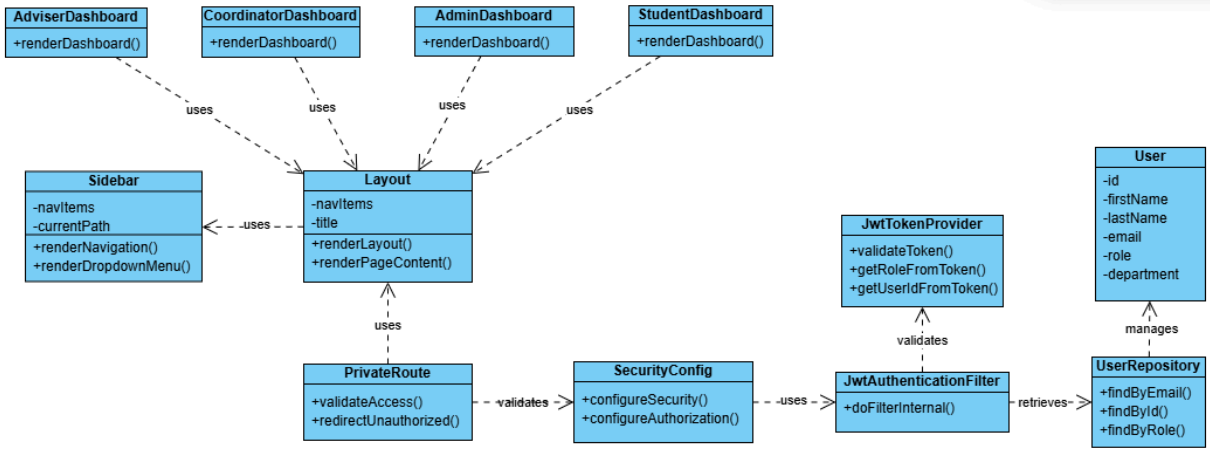
- Description and purpose
  - Represents the authenticated system user entity containing account credentials, assigned system roles, and user profile information required for access control and authorization.
- Component type or format
  - Entity Class (entity/User.java)

o UserDTO

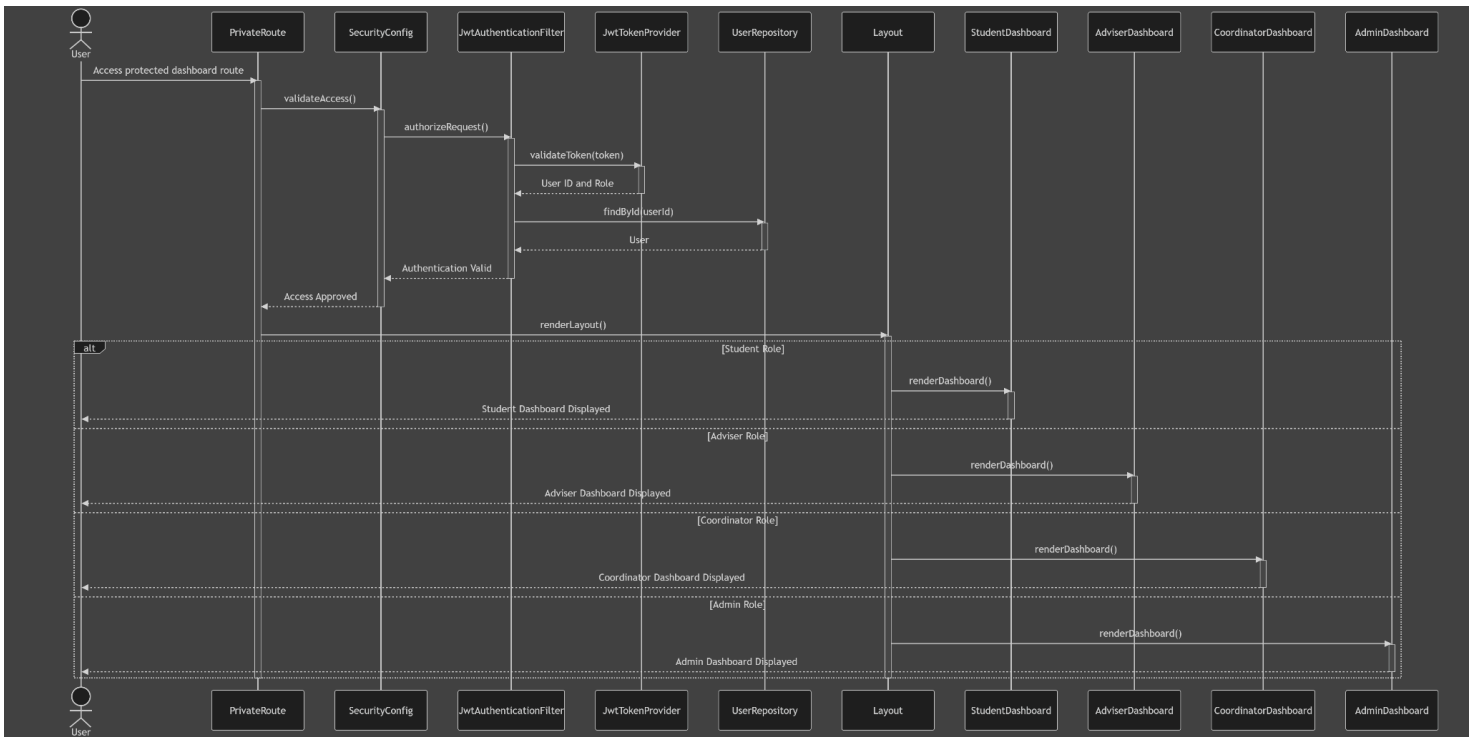
- Description and purpose
  - Transfers user-related data between backend services and the frontend. It ensures that only necessary and safe user information is exposed, such as user role and profile details, while protecting sensitive data.
- Component type or format
  - Data Transfer Object (dto/UserDTO.java)

- Object-Oriented Components
  - Class Diagram

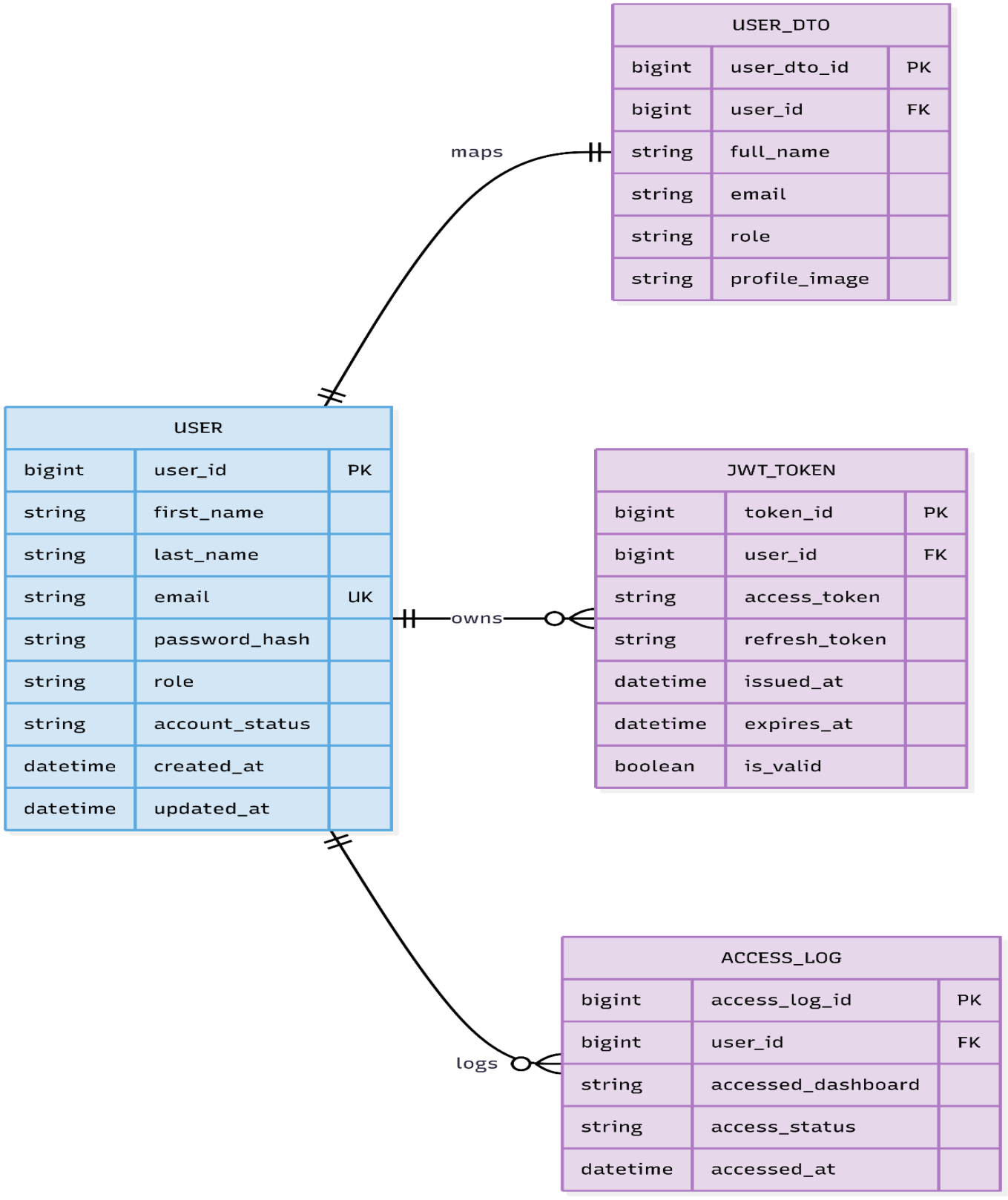
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- Sequence Diagram

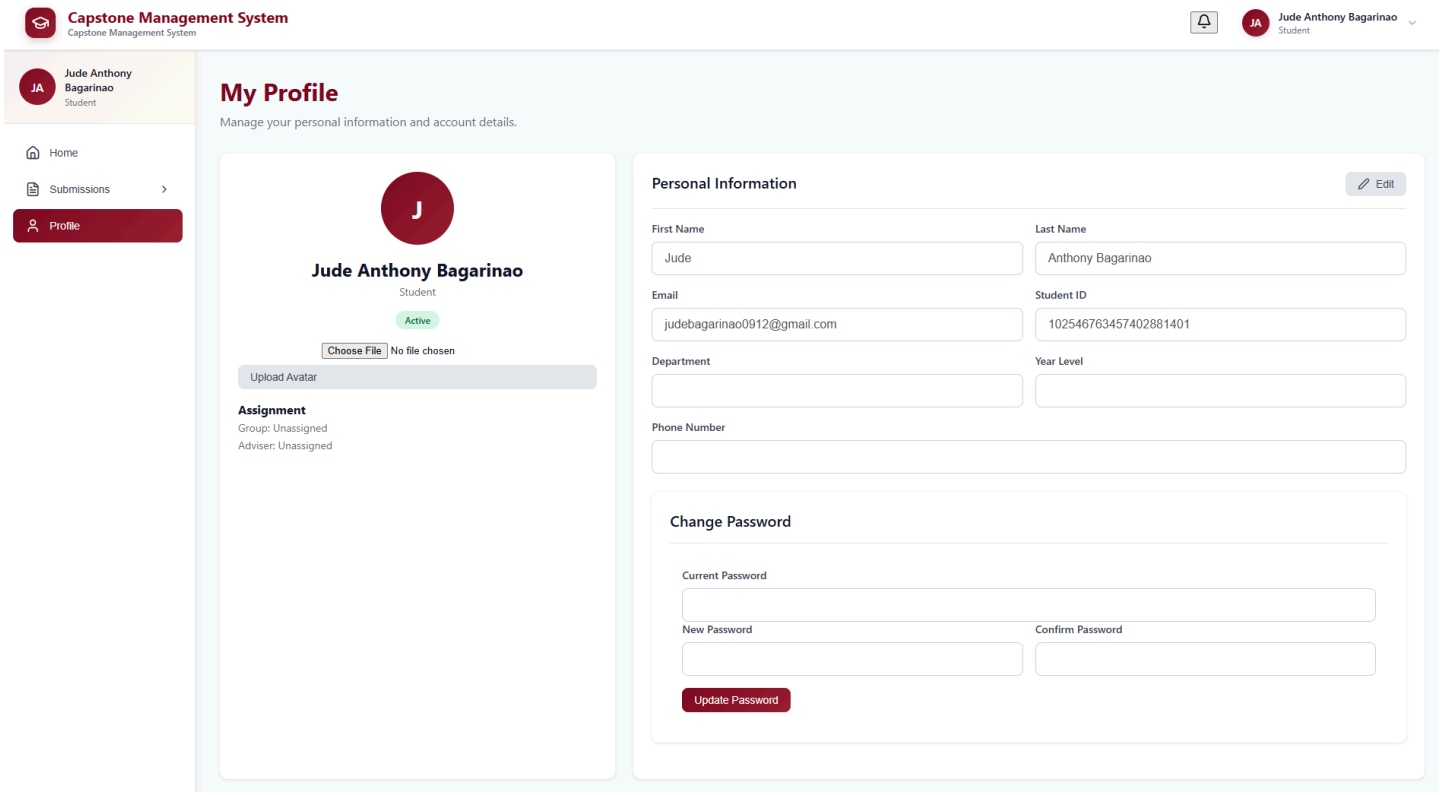


- Data Design
  - ERD or schema

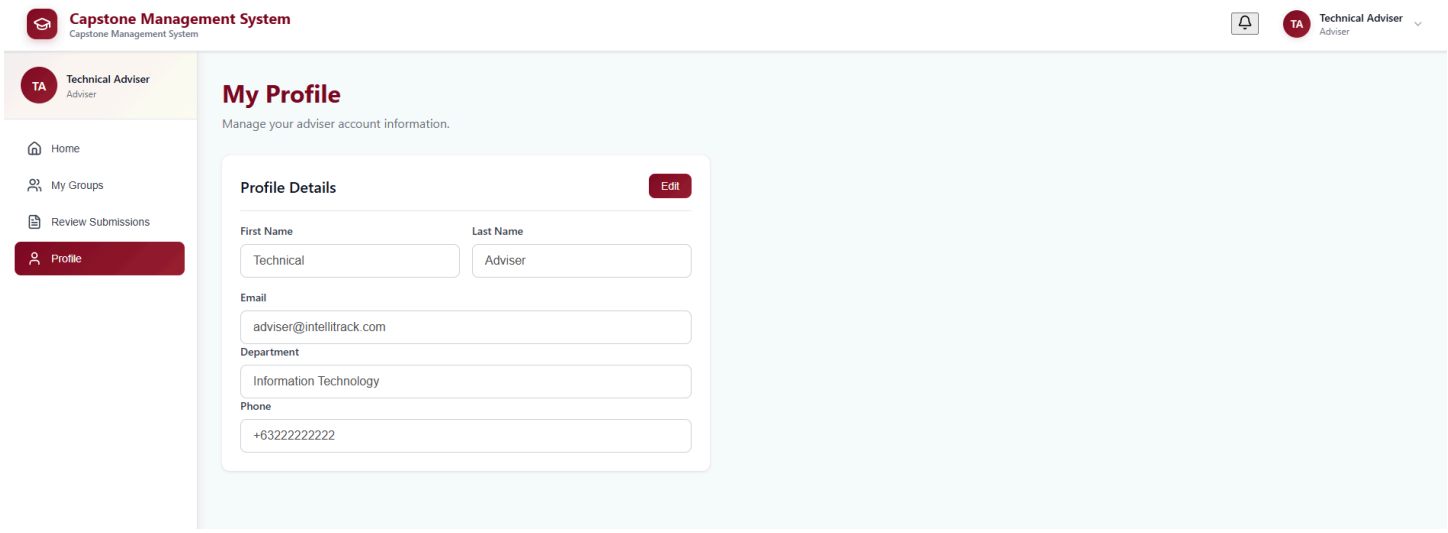

USER_DTO
bigint user_dto_id PK
bigint user_id FK
string full_name
string email
string role
string profile_image
maps
USER
bigint user_id PK
string first_name
string last_name
string email UK
string password_hash
string role
string account_status
datetime created_at
datetime updated_at
owns
JWT_TOKEN
bigint token_id PK
bigint user_id FK
string access_token
string refresh_token
datetime issued_at
datetime expires_at
boolean is_valid
logs
ACCESS_LOG
bigint access_log_id PK
bigint user_id FK
string accessed_dashboard
string access_status
datetime accessed_at

1.3 Profile Management

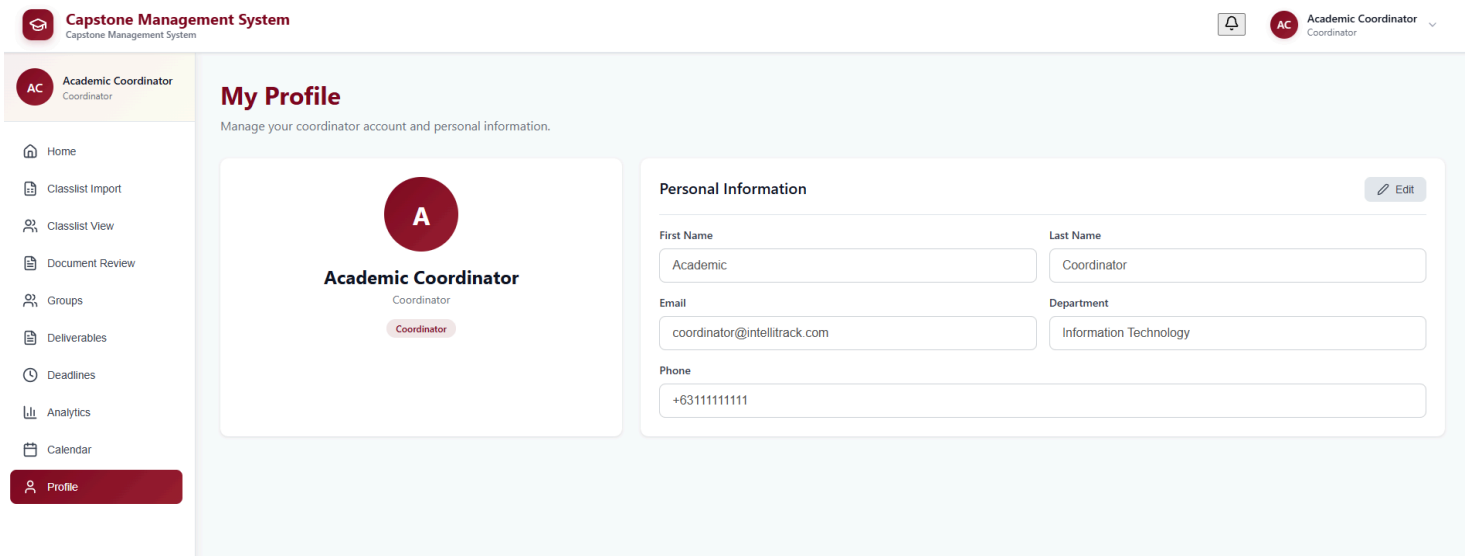
- User Interface Design
    - Student



    - Advisor



    - Coordinator



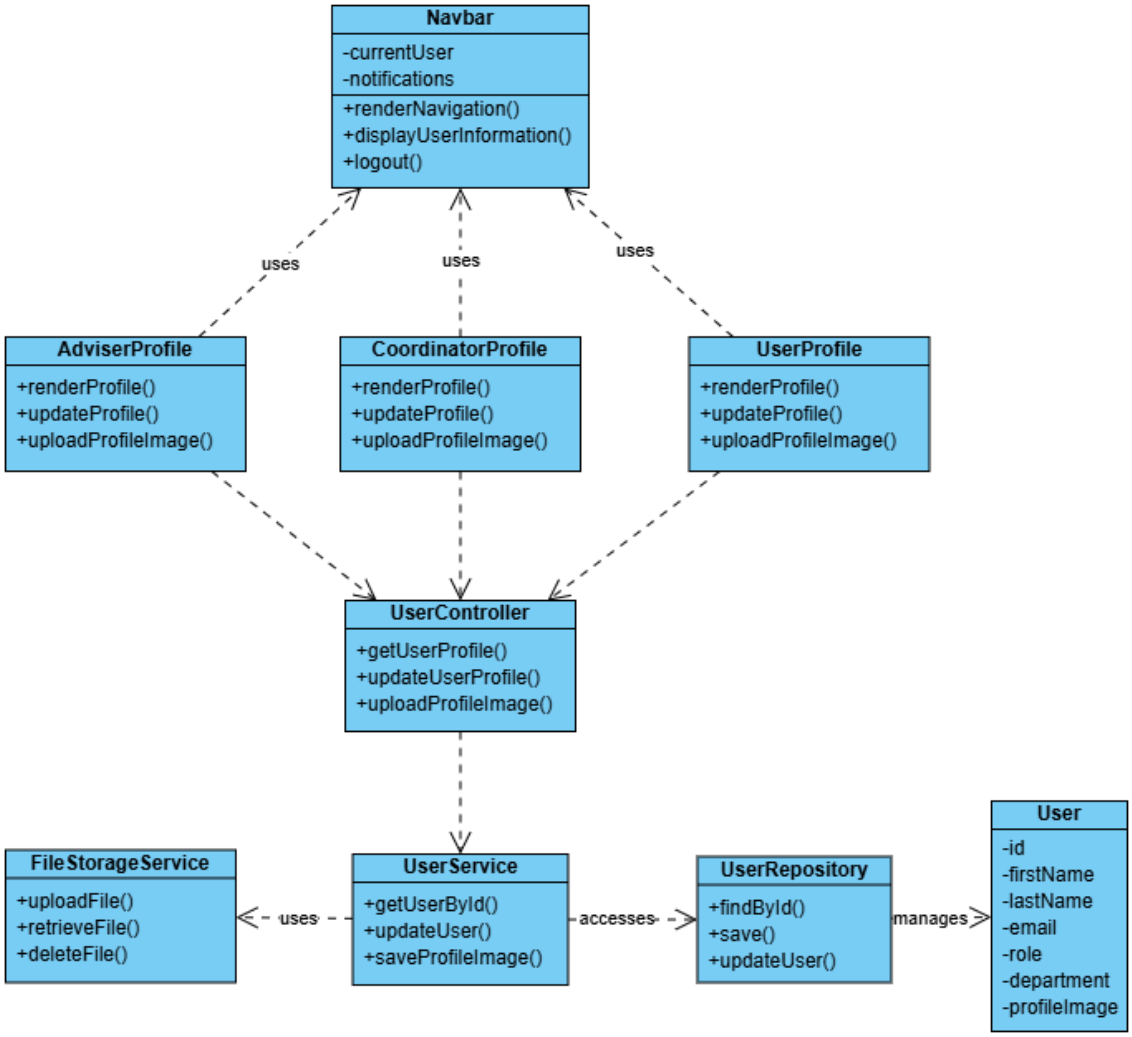
- Front-end component(s)
    - **AdviserProfile**
        - Description and purpose
            - Allows adviser users to view and update their personal account information including first name, last name, department, and contact details. It supports profile editing and profile data persistence within the system.
        - Component type or format
            - React Profile Component (pages/adviser/AdviserProfile.jsx)
    - **CoordinatorProfile**
        - Component Name
            - Description and purpose
                - Allows coordinator users to manage and update their personal profile information including account and contact details used within the system.
            - Component type or format
                - React Profile Component (pages/coordinator/CoordinatorProfile.jsx)
    - **UserProfile**
        - Component Name
            - Description and purpose
                - Allows coordinator users to manage and update their personal profile iProvides a reusable profile interface structure for authenticated users to access and manage their profile-related information within the system.
            - Component type or format

- React Profile Component (components/UserProfile.jsx)

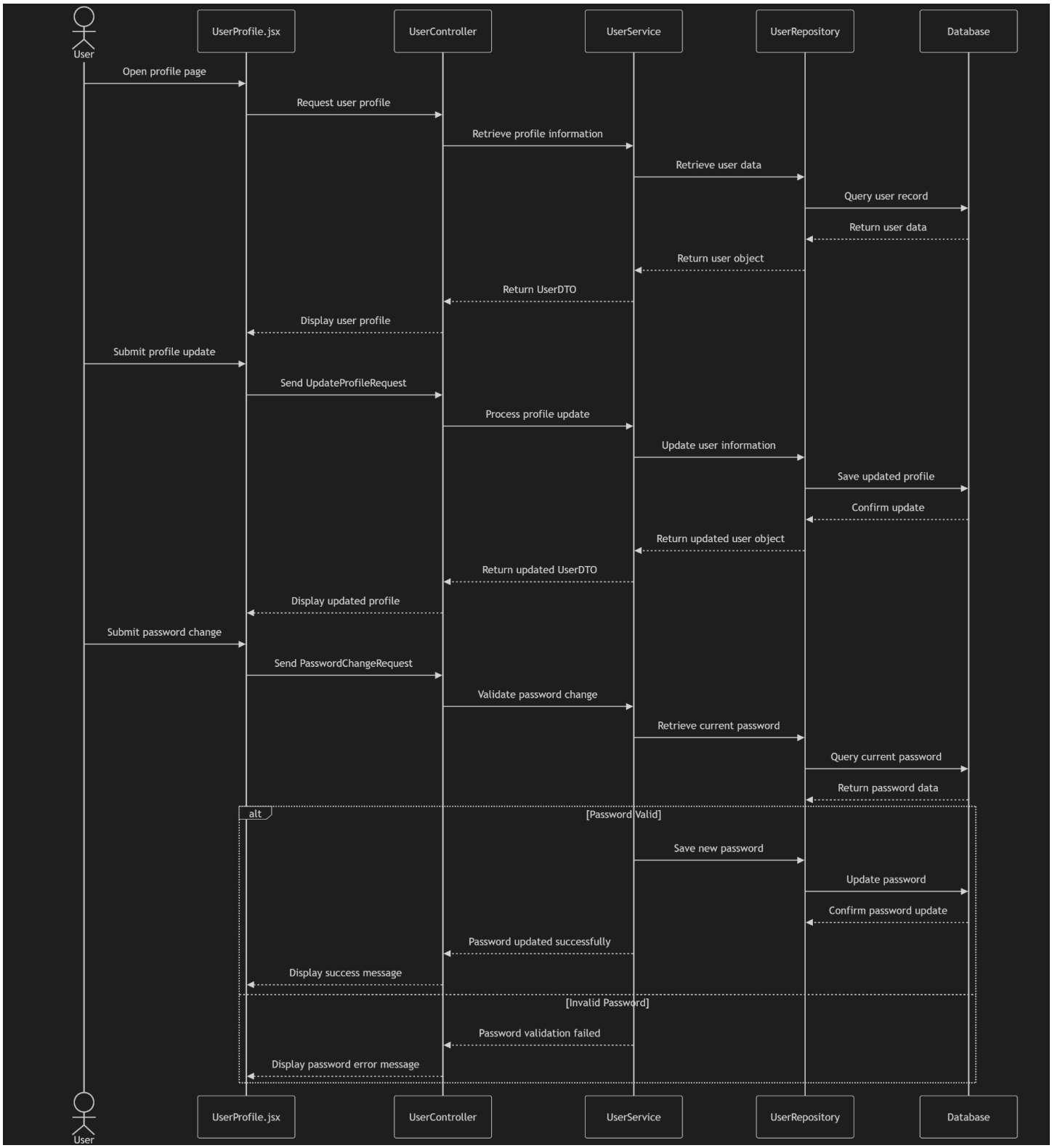
- Back-end component(s)
  - **UserService**
    - Description and purpose
      - Handles the business logic related to user profile retrieval and profile update operations. It processes user information changes and ensures profile data consistency within the system.
    - Component type or format
      - Service Class (service/UserService.java)
  - **UserRepository**
    - Description and purpose
      - Retrieves and updates user profile information stored in the database including personal details, account credentials, and assigned user roles.
    - Component type or format
      - Repository Interface (repository/UserRepository.java)
  - **User**
    - Description and purpose
      - Represents the system user entity containing profile information such as personal details, email credentials, department information, contact information, and assigned system roles.
    - Component type or format
      - Entity Class (entity/User.java)
  - **FileStorageService**
    - Description and purpose
      - Handles file storage operations related to user-uploaded profile resources and manages storage using local storage or cloud-based storage integration.
    - Component type or format
      - Service Class (service/FileStorageService.java)

- Object-Oriented Components
  - Class Diagram

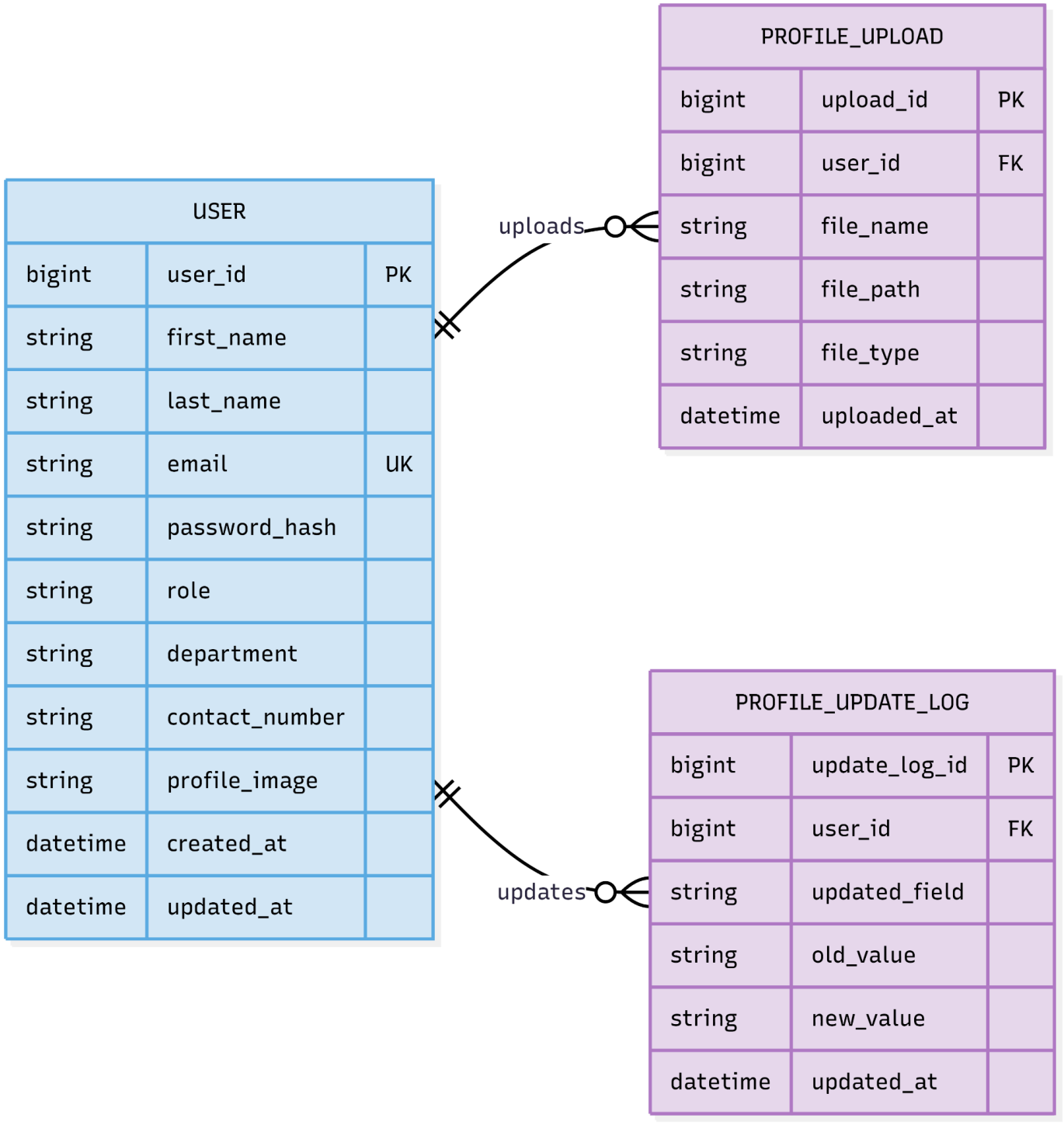
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o Sequence Diagram

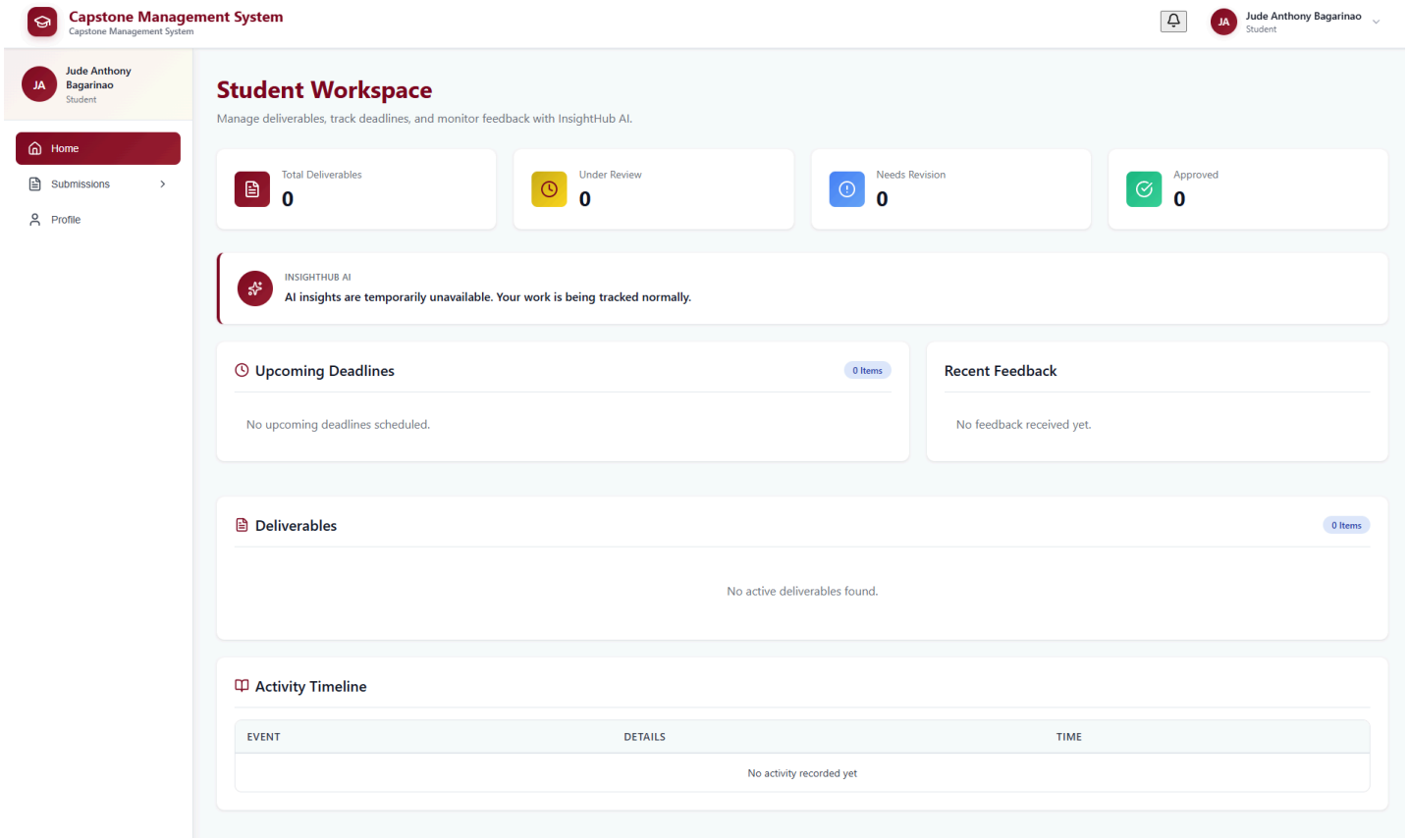

User
UserProfile.jsx
UserController
UserService
UserRepository
Database
Open profile page
Request user profile
Retrieve profile information
Retrieve user data
Query user record
Return user data
Return user object
Return UserDTO
Display user profile
Submit profile update
Send UpdateProfileRequest
Process profile update
Update user information
Save updated profile
Confirm update
Return updated user object
Return updated UserDTO
Display updated profile
Submit password change
Send PasswordChangeRequest
Validate password change
Retrieve current password
Query current password
Return password data
alt
[Password Valid]
Save new password
Update password
Confirm password update
Password updated successfully
Display success message
[Invalid Password]
Password validation failed
Display password error message
User
UserProfile.jsx
UserController
UserService
UserRepository
Database

- Data Design
    - ERD or schema

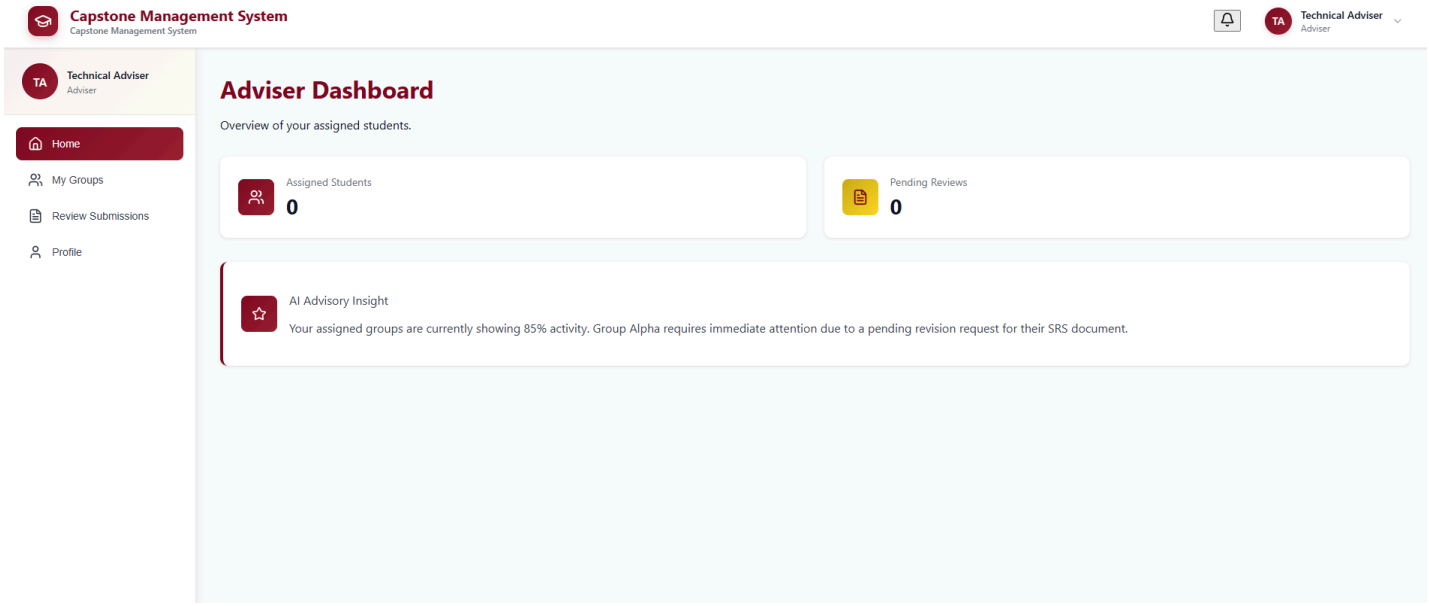


1.4 Home Page

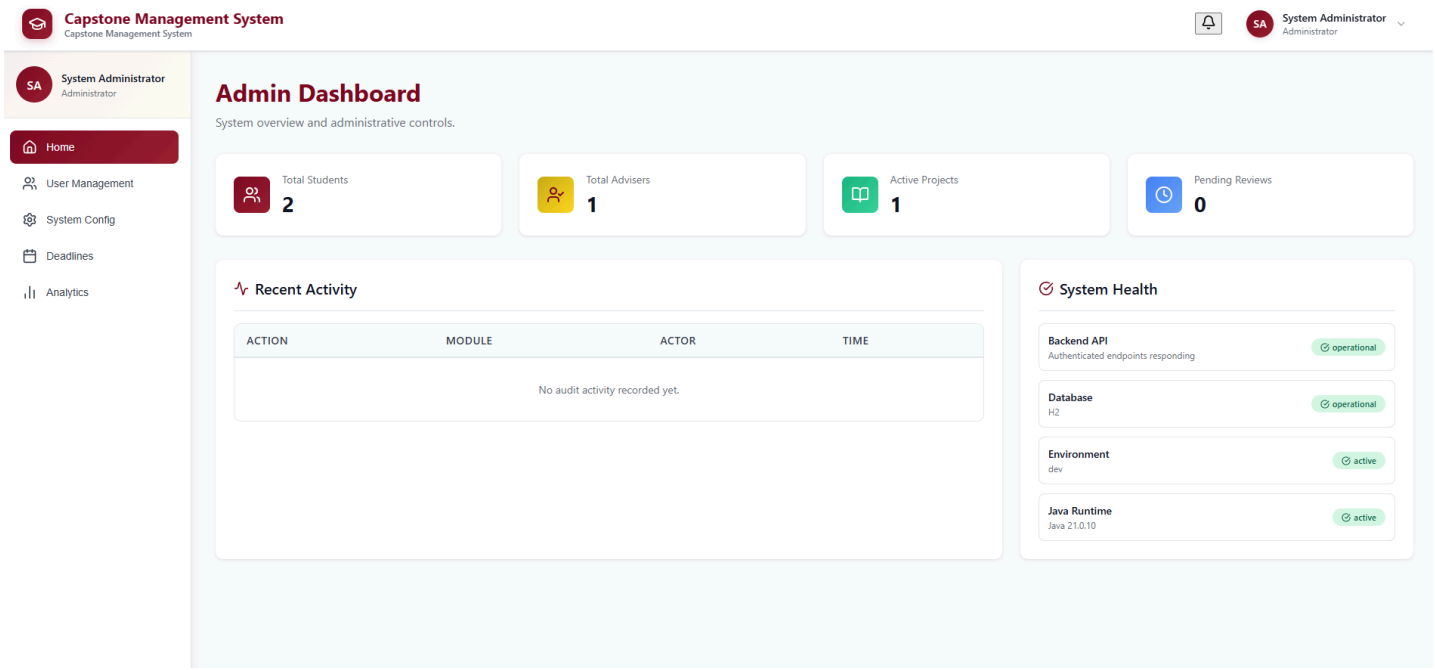
- User Interface Design
    - Student



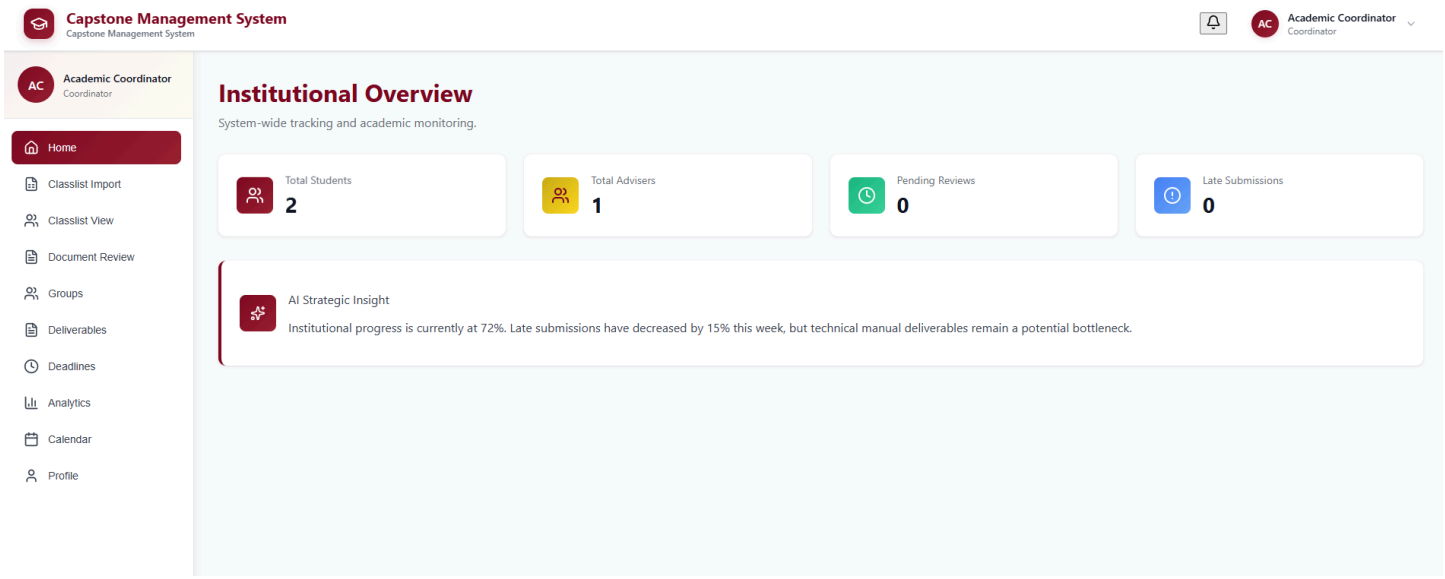
    - Advisor



o Administrator



o Coordinator



- Front-end component(s)
  - **AdminHome**
    - Description and purpose
      - Provides the administrator dashboard interface displaying system-wide information, analytics summaries, and access to administrative management modules.
    - Component type or format
      - React Dashboard Component (pages/admin/AdminHome.jsx)
  - **AdviserHome**
    - Description and purpose
      - Provides advisers with an overview of assigned students, pending submission reviews, and AI-generated advisory insights related to student progress and submission monitoring.
    - Component type or format
      - React Dashboard Component (pages/adviser/AdviserHome.jsx)
  - **CoordinatorHome**
    - Description and purpose
      - Provides coordinators with institutional monitoring information including total students, advisers, pending submissions, late submissions, and AI-generated strategic insights.
    - Component type or format
      - React Dashboard Component (pages/coordinator/CoordinatorHome.jsx)

- **StudentHome**
  - Description and purpose
    - Provides students with dashboard access to submission-related activities, active deliverables, and project monitoring information within the system.
  - Component type or format
    - React Dashboard Component (pages/student/StudentHome.jsx)
- **Sidebar**
  - Description and purpose
    - Displays the role-based navigation menu of the system. It dynamically renders navigation items and dropdown modules depending on the authenticated user role and available permissions.
  - Component type or format
    - React Navigation Component (components/Sidebar.jsx)
- **Navbar**
  - Description and purpose
    - Displays authenticated user information within the system interface, including user name, role, profile access options, notification access, account settings access, and logout functionality.
  - Component type or format
    - React Navigation Component (components/Navbar.jsx)
- **Layout**
  - Description and purpose
    - Acts as the main structural wrapper of the frontend application. It combines the Navbar, Sidebar, and page content area while maintaining consistent dashboard layout and navigation across user roles.
  - Component type or format
    - React Layout Wrapper (components/Layout.jsx)
- **MetricCard**
  - Description and purpose

- Displays summarized dashboard metrics such as submission counts, analytics values, and monitoring statistics in a compact card-based format for quick visualization.

- Component type or format

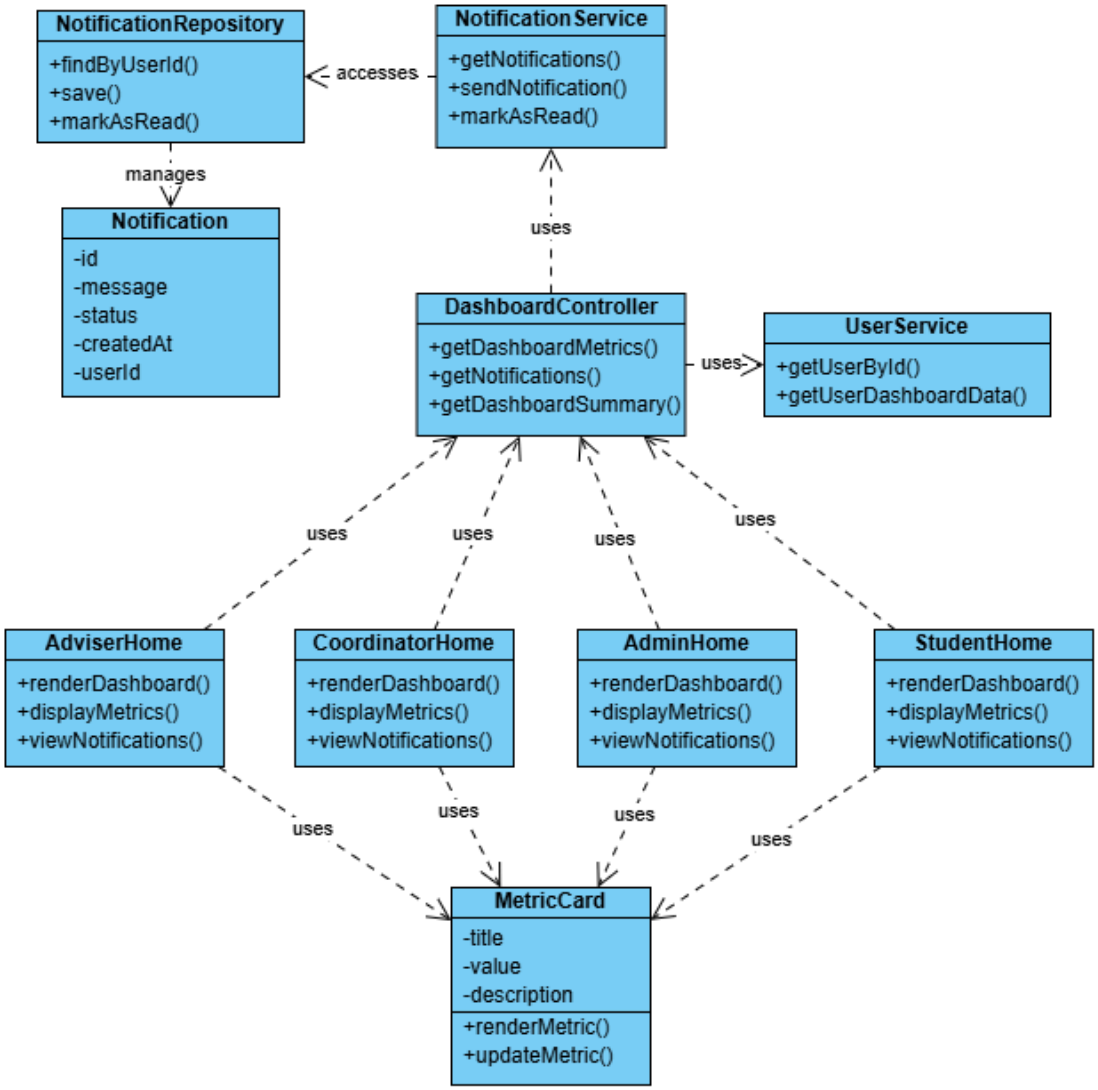
- React Dashboard Component (components/MetricCard.jsx)

- Back-end component(s)
  - **DashboardController**
    - Description and purpose
      - HActs as the API entry point for retrieving dashboard-related information and analytics summaries for different authenticated user roles within the system.
    - Component type or format
      - REST Controller (controller/DashboardController.java)
- **UserService**
  - Description and purpose
    - Retrieves user-related dashboard information and manages user-specific data required for dashboard presentation and monitoring.
  - Component type or format
    - Service Class (service/UserService.java)
- **NotificationService**
  - Description and purpose
    - Handles the creation, retrieval, and management of dashboard notifications and alerts for authenticated system users.
  - Component type or format
    - Service Class (service/NotificationService.java)
- **NotificationRepository**
  - Description and purpose
    - Retrieves and stores notification records associated with authenticated users and dashboard activities within the database.
  - Component type or format
    - Repository Interface (repository/NotificationRepository.java)
- **Notification**
  - Description and purpose
    - Represents notification records generated by the system for alerts, reminders, dashboard updates, and submission-related activities associated with authenticated users.

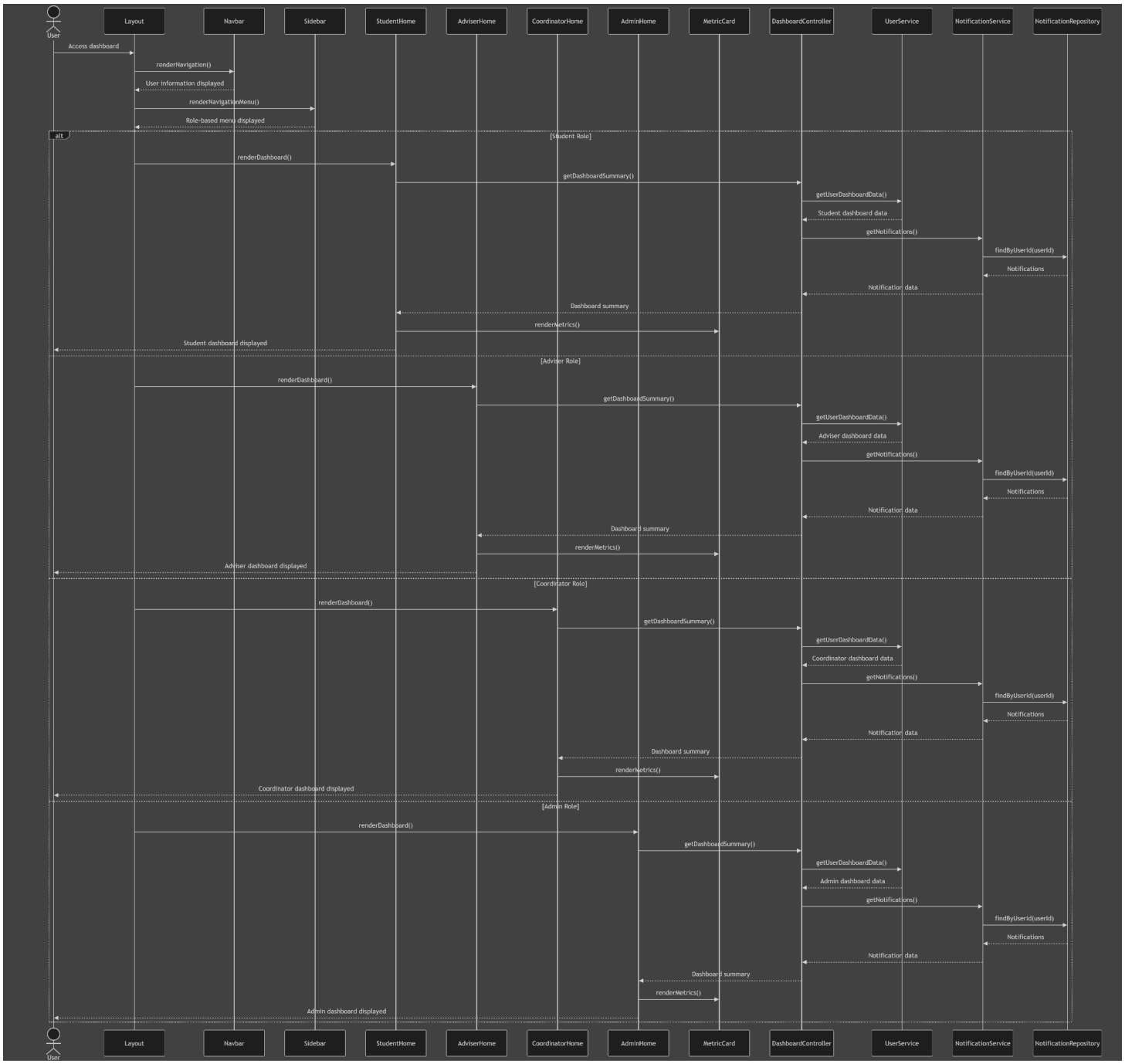
    - Component type or format
        - Entity Class (entity/Notification.java)

- Object-Oriented Components
    - Class Diagram

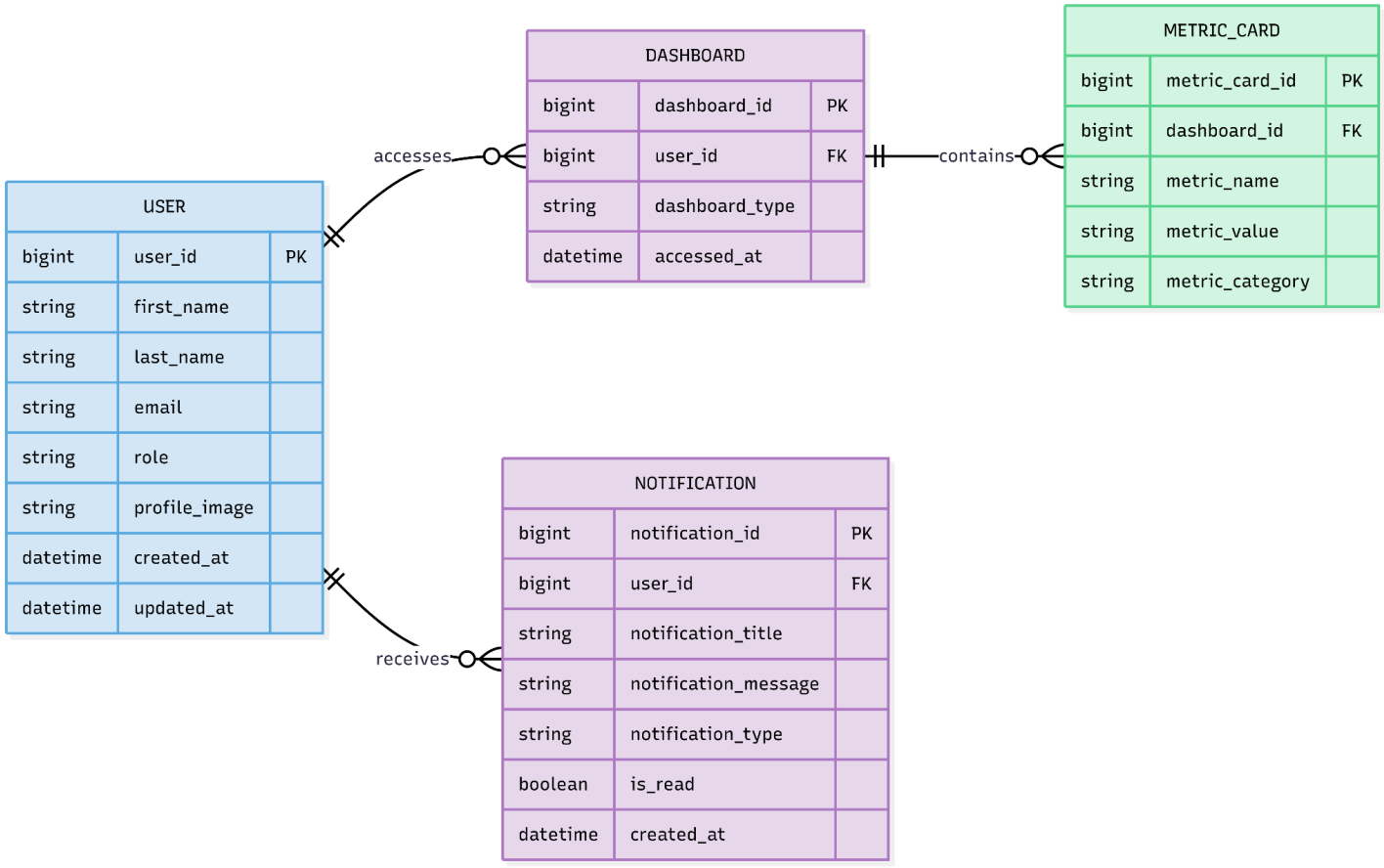
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- Sequence Diagram



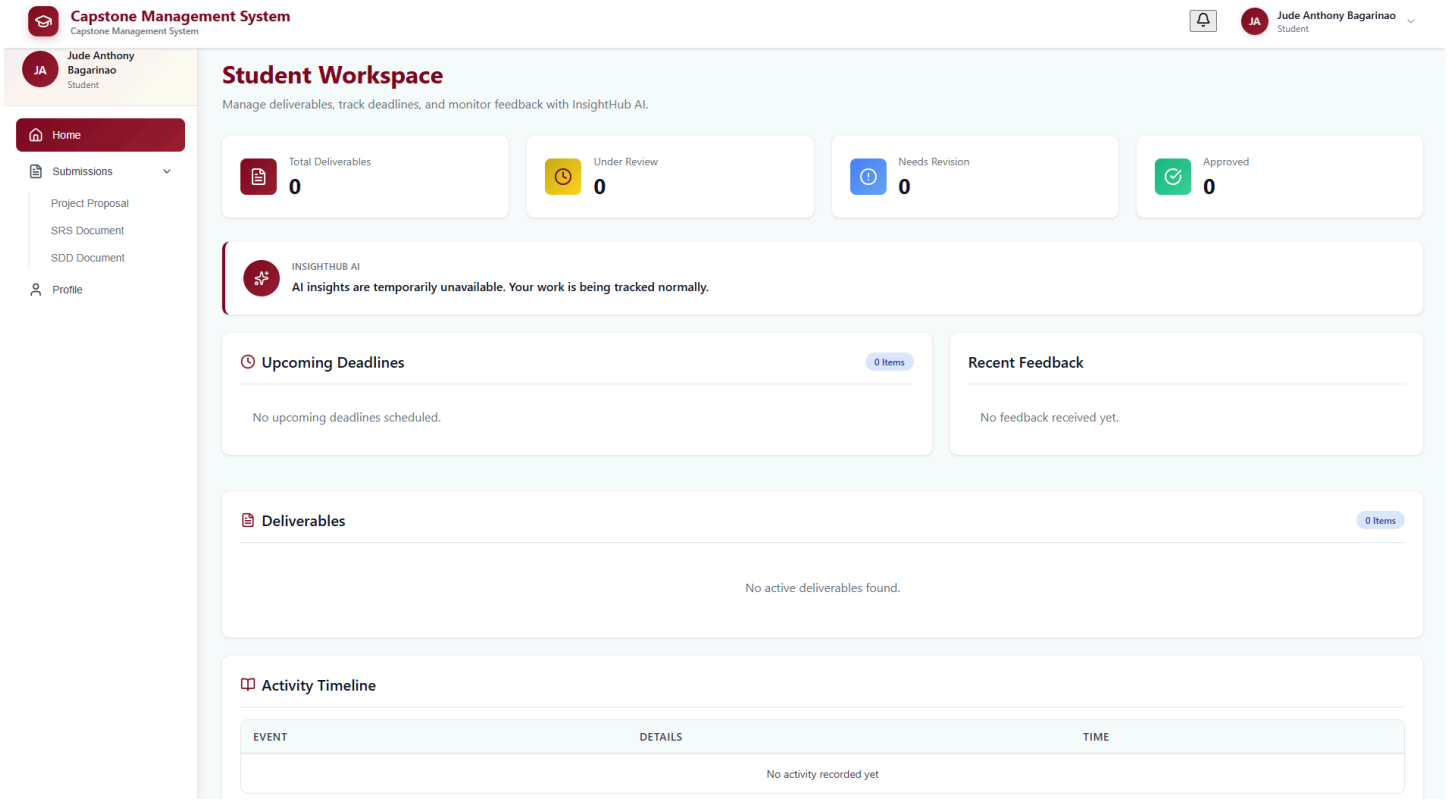
- Data Design
    - ERD or schema



Module 2: Submission Tracker

2.1 Deadline and AI Risk Monitoring

- User Interface Design



- Front-end component(s)
  - **SubmissionTrackerDashboard**
    - Description and purpose
      - Provides a centralized dashboard for monitoring active submission deadlines, countdown windows, AI-generated risk levels, and deliverable tracking information for student groups.
    - Component type or format
      - React Dashboard Component (components/SubmissionTrackerDashboard.jsx)
  - **StatusMonitoringPanel**
    - Description and purpose
      - Displays the current submission status of deliverables including deadlines, countdown timers, submission timestamps, revision counts, and monitoring indicators for each milestone.
    - Component type or format
      - React Monitoring Component (components/StatusMonitoringPanel.jsx)

- Back-end component(s)
  - **DeadlineRepository**

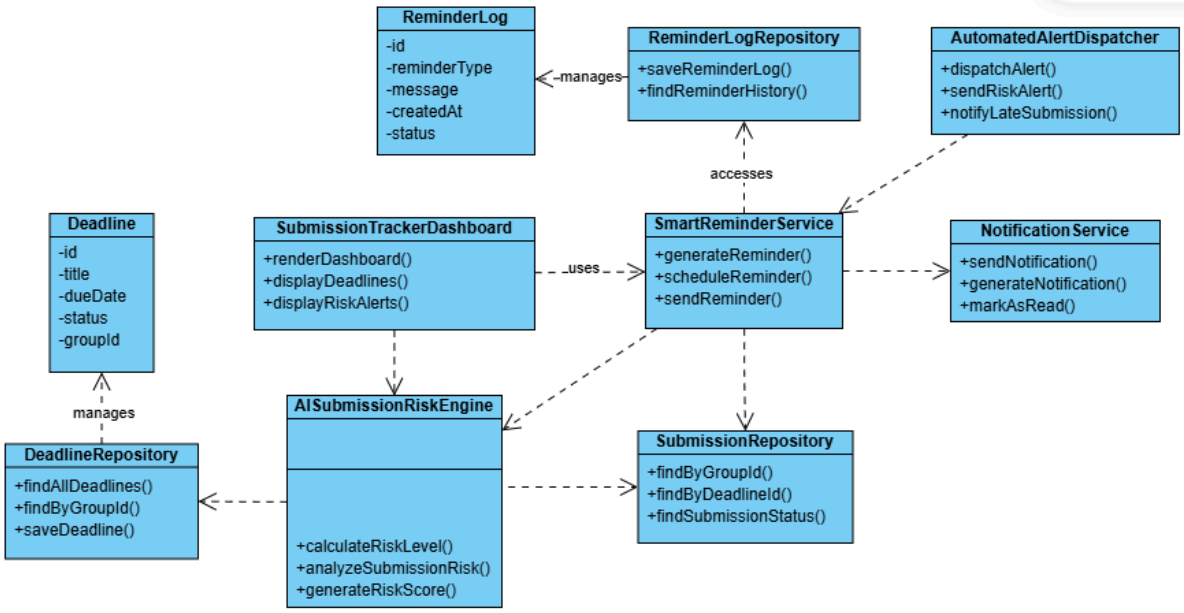
- Description and purpose
  - Retrieves and manages deliverable deadline records stored in the database including due dates and deadline schedules used for submission monitoring.
- Component type or format
  - Repository Interface (repository/DeadlineRepository.java)

- **Deadline**
  - Description and purpose
    - Represents deliverable deadline information including due dates and milestone schedules used for submission monitoring and countdown tracking.
  - Component type or format
    - Entity Class (entity/Deadline.java)
- **AISubmissionRiskEngine**
  - Description and purpose
    - Analyzes submission-related factors such as remaining hours, submission status, revision count, and previous late submissions to generate AI-assisted risk scores and risk level assessments for deliverables.
  - Component type or format
    - Service Class (service/AISubmissionRiskEngine.java)
- **SmartReminderService**
  - Description and purpose
    - Generates AI-assisted reminder messages based on submission risk levels, deadline proximity, and deliverable monitoring information.
  - Component type or format
    - Service Class (service/SmartReminderService.java)
- **AutomatedAlertDispatcher**
  - Description and purpose
    - Automatically dispatches scheduled submission reminders and monitoring alerts for approaching deadlines and pending deliverables.
  - Component type or format
    - Scheduled Service Component (service/AutomatedAlertDispatcher.java)
- **ReminderLogRepository**
  - Description and purpose
    - Stores and retrieves records of dispatched reminder notifications and automated alert activities generated by the system.
  - Component type or format

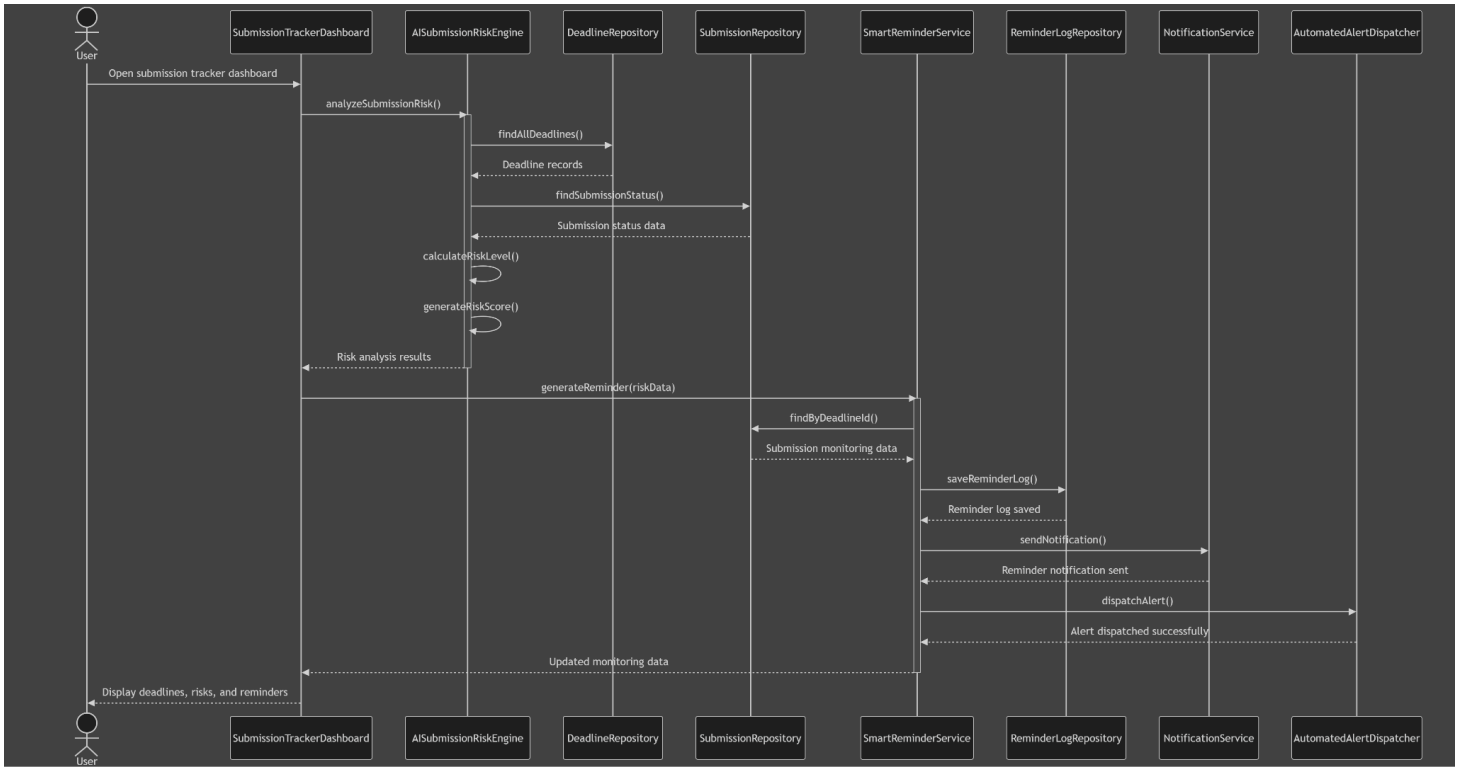
        - Repository Interface (repository/ReminderLogRepository.java)
    - **ReminderLog**
        - Description and purpose
            - Represents reminder notification records including dispatched messages, delivery channels, associated project groups, and deliverable references.
        - Component type or format
            - Entity Class (entity/ReminderLog.java)
    - **NotificationService**
        - Description and purpose
            - Handles the generation and delivery of notification alerts related to submission deadlines, reminders, and monitoring activities within the system.
        - Component type or format
        - Service Class (service/NotificationService.java)

- Object-Oriented Components
    - Class Diagram

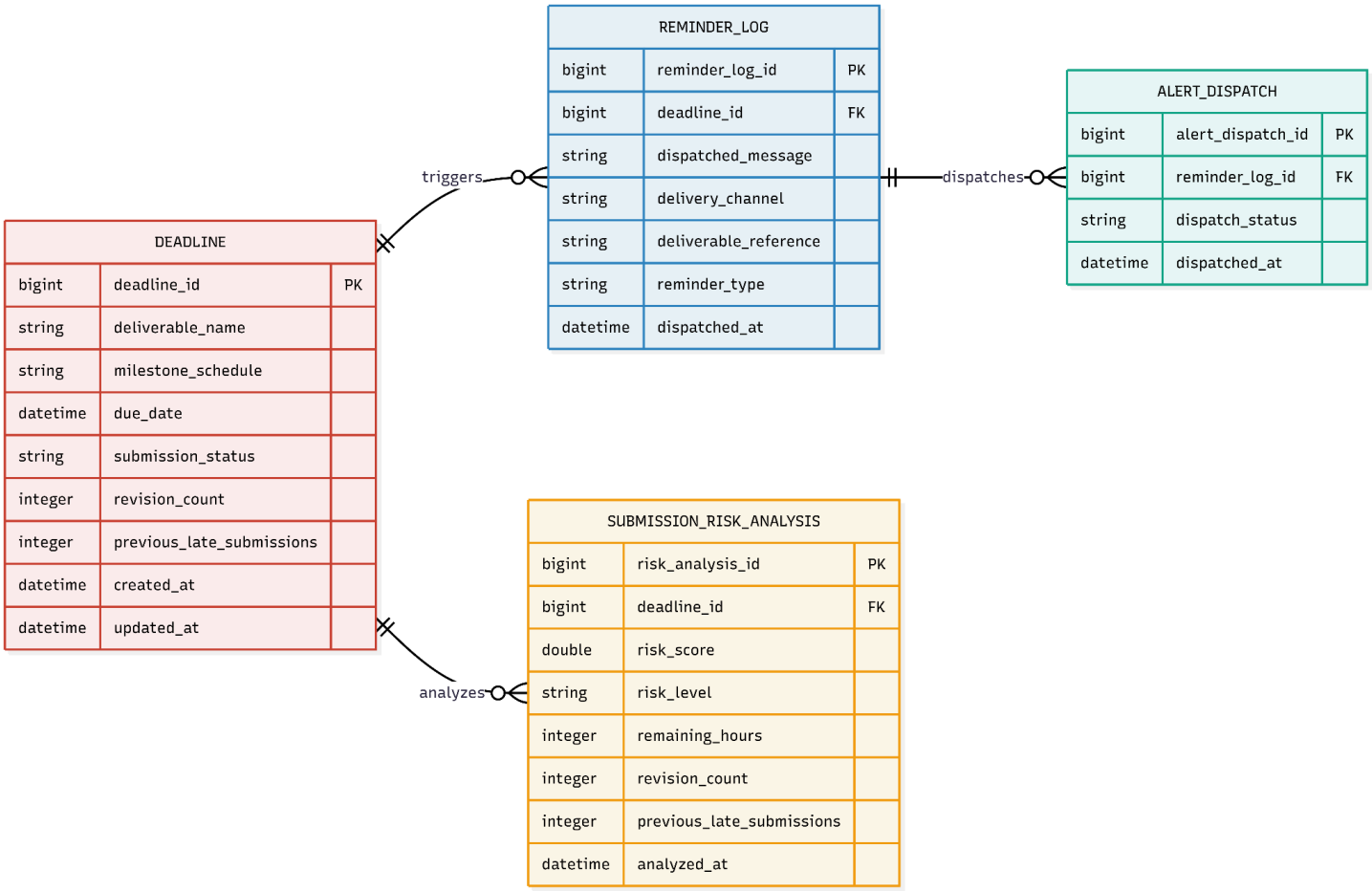
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- Sequence Diagram

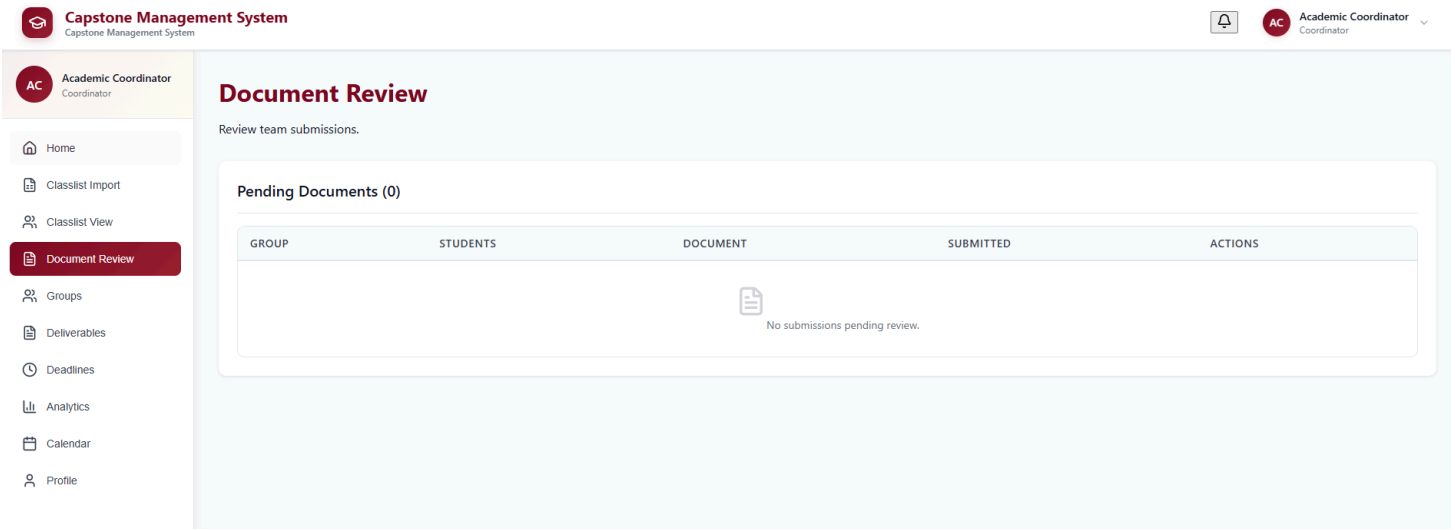


- Data Design
    - ERD or schema



2.2 Status Monitoring

- User Interface Design



- Front-end component(s)
    - **StatusMonitoringPanel**
        - Description and purpose
            - Displays the current monitoring status of deliverables including submission progress, deadlines, revision counts, submission timestamps, and status indicators for each milestone.
        - Component type or format
            - **React Monitoring Component (components/StatusMonitoringPanel.jsx)**
    - **SubmissionTrackingDashboard**
        - Description and purpose
            - Provides a dashboard interface for tracking submission progress, monitoring completion rates, visualizing submission analytics, and reviewing overall deliverable status across project groups.
        - Component type or format
            - **React Dashboard Component (components/SubmissionTrackingDashboard.jsx)**
    - **ProgressList**
        - Description and purpose
            - Displays project group progress information including completion percentages, submitted deliverables, and overall submission progress visualization.
        - Component type or format
            - **React Progress Monitoring Component (components/ProgressList.jsx)**
    - **ActivityFeed**
        - Description and purpose
            - Displays recent submission-related activities, status updates, and monitoring events associated with project deliverables and submission tracking.
        - Component type or format
            - **React Activity Monitoring Component (components/ActivityFeed.jsx)**
- Back-end component(s)
    - **SubmissionMetricsEngine**
        - Description and purpose
            - Processes submission-related records and computes monitoring analytics including completion percentages, submission rates, pending submissions,

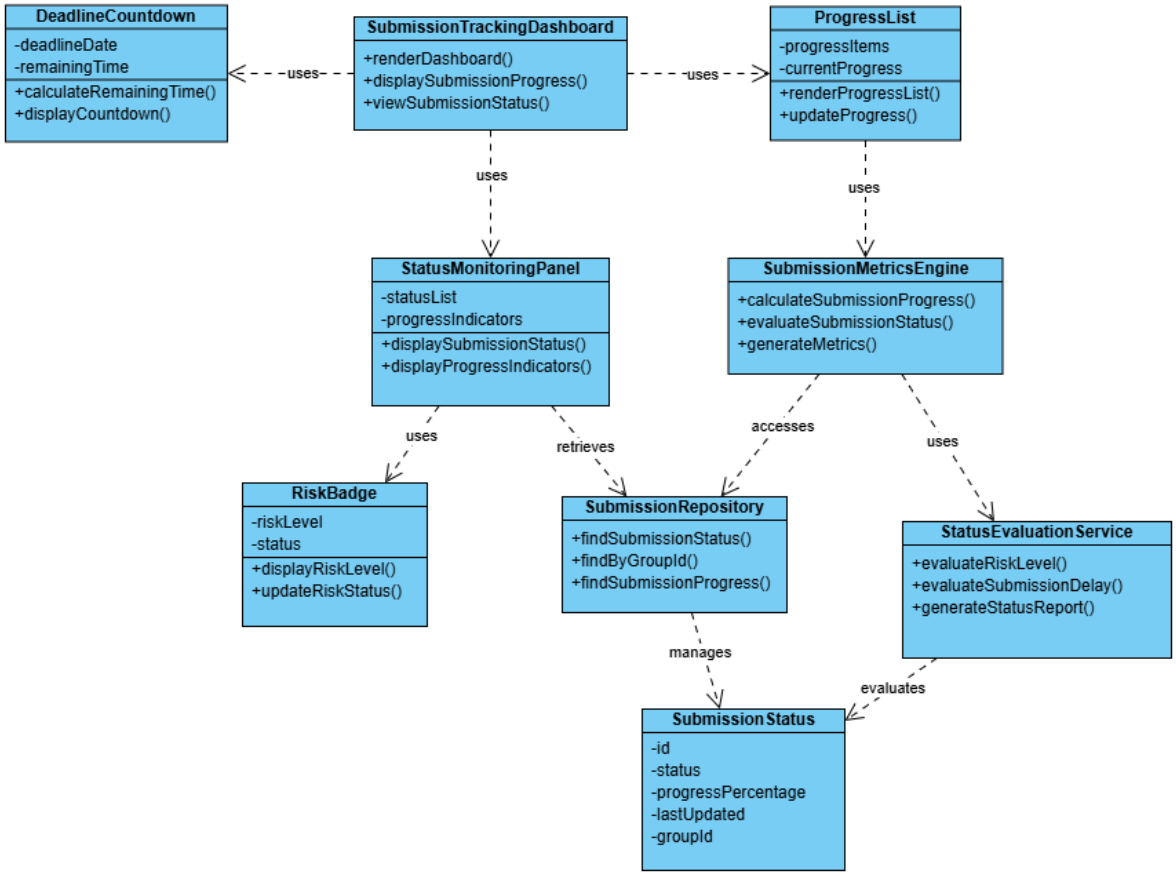
and late submission statistics used for status tracking dashboards.

- Component type or format
  - Service Class (service/SubmissionMetricsEngine.java)

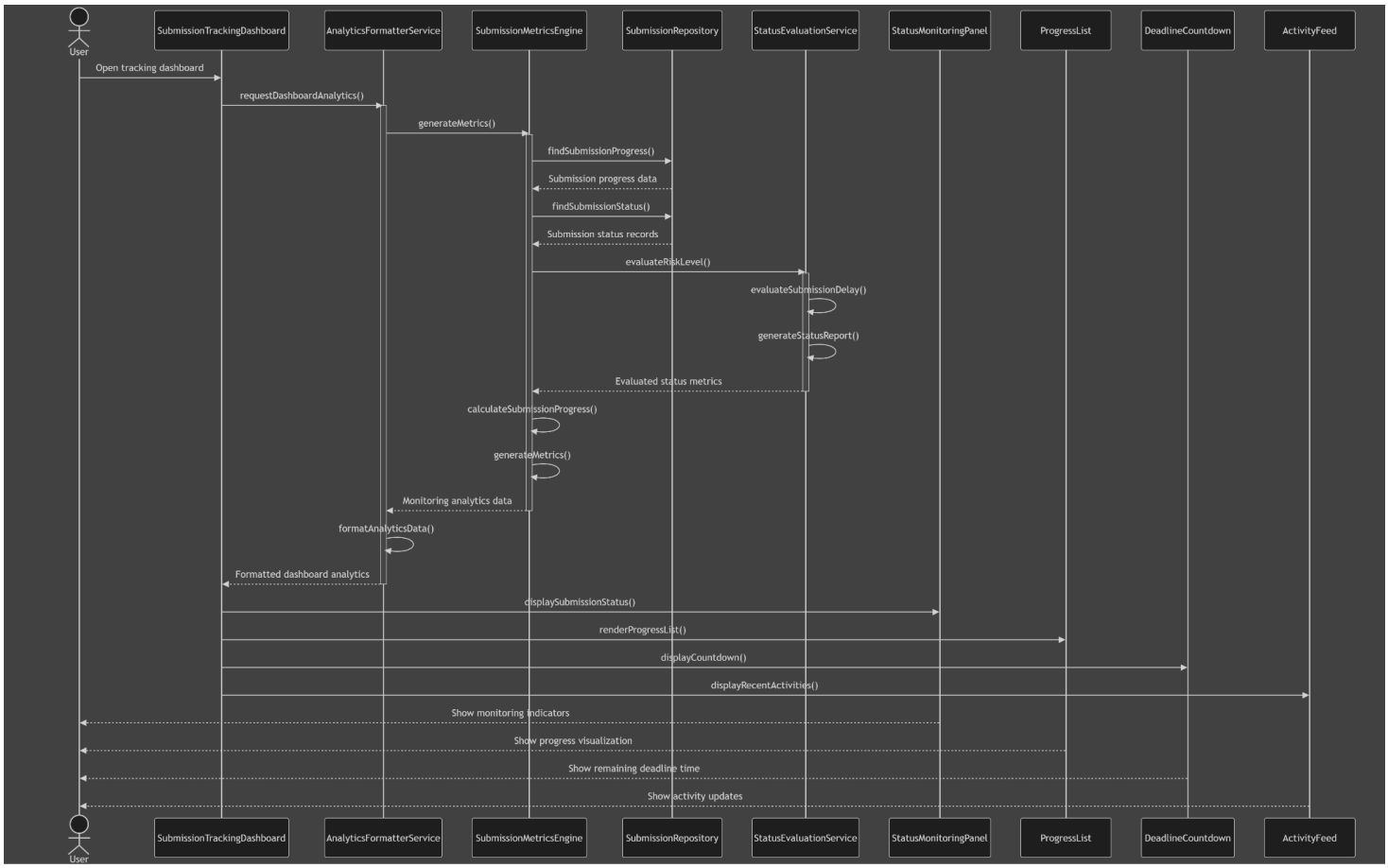
- **SubmissionStatus**
  - Description and purpose
    - Defines the available submission status classifications used within the system including pending, submitted, updated, and late submission states for deliverable monitoring.
  - Component type or format
    - Enumeration (entity/SubmissionStatus.java)
- **StatusEvaluationService**
  - Description and purpose
    - Evaluates and determines the current submission status of deliverables based on submission timestamps, deadlines, and monitoring conditions.
  - Component type or format
    - Service Class (service/StatusEvaluationService.java)
- **AnalyticsFormatterService**
  - Description and purpose
    - Formats processed submission monitoring data into structured dashboard analytics representations used for frontend visualization and reporting components.
  - Component type or format
    - Service Class (service/AnalyticsFormatterService.java)

- Object-Oriented Components
  - Class Diagram

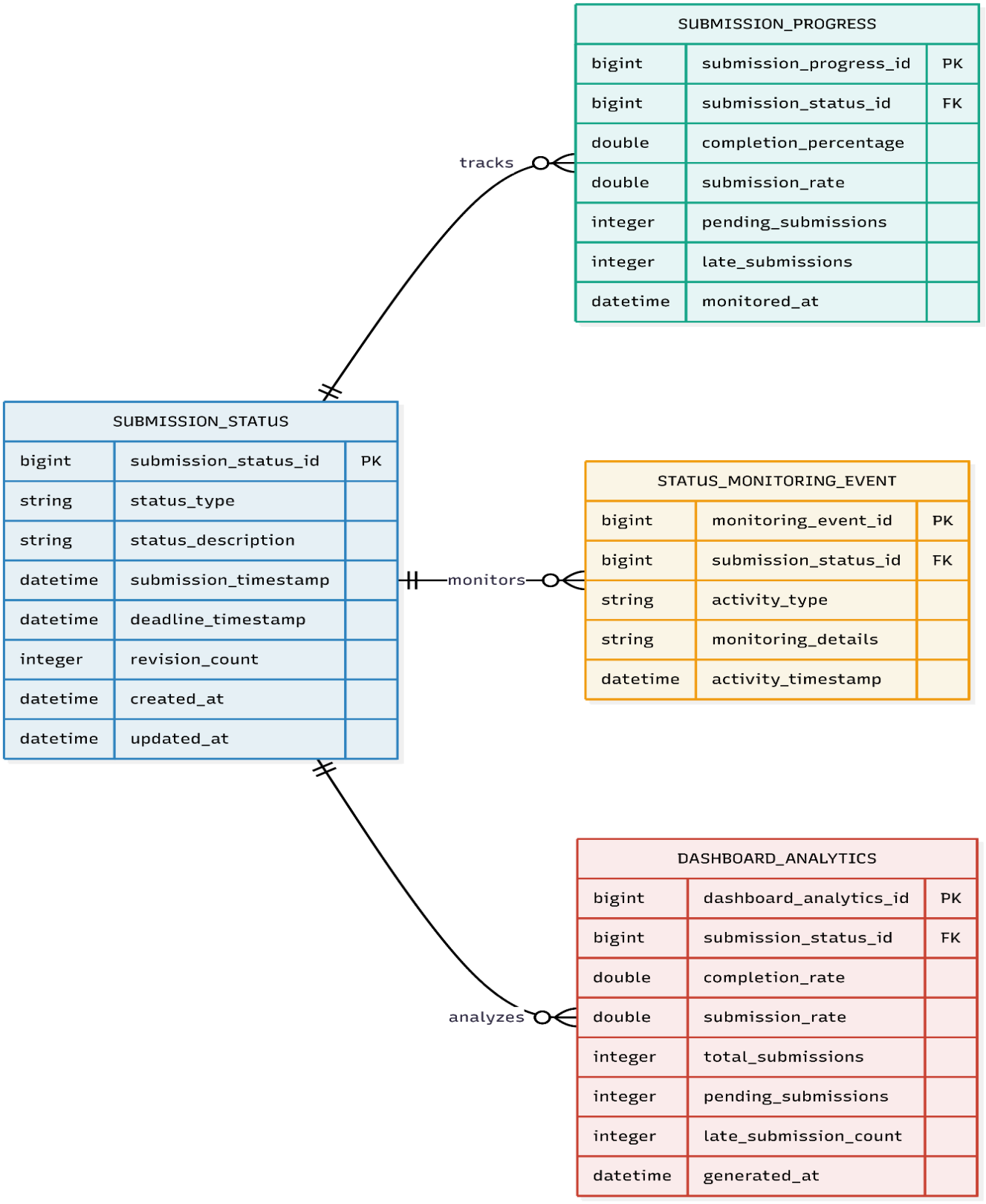
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- Sequence Diagram

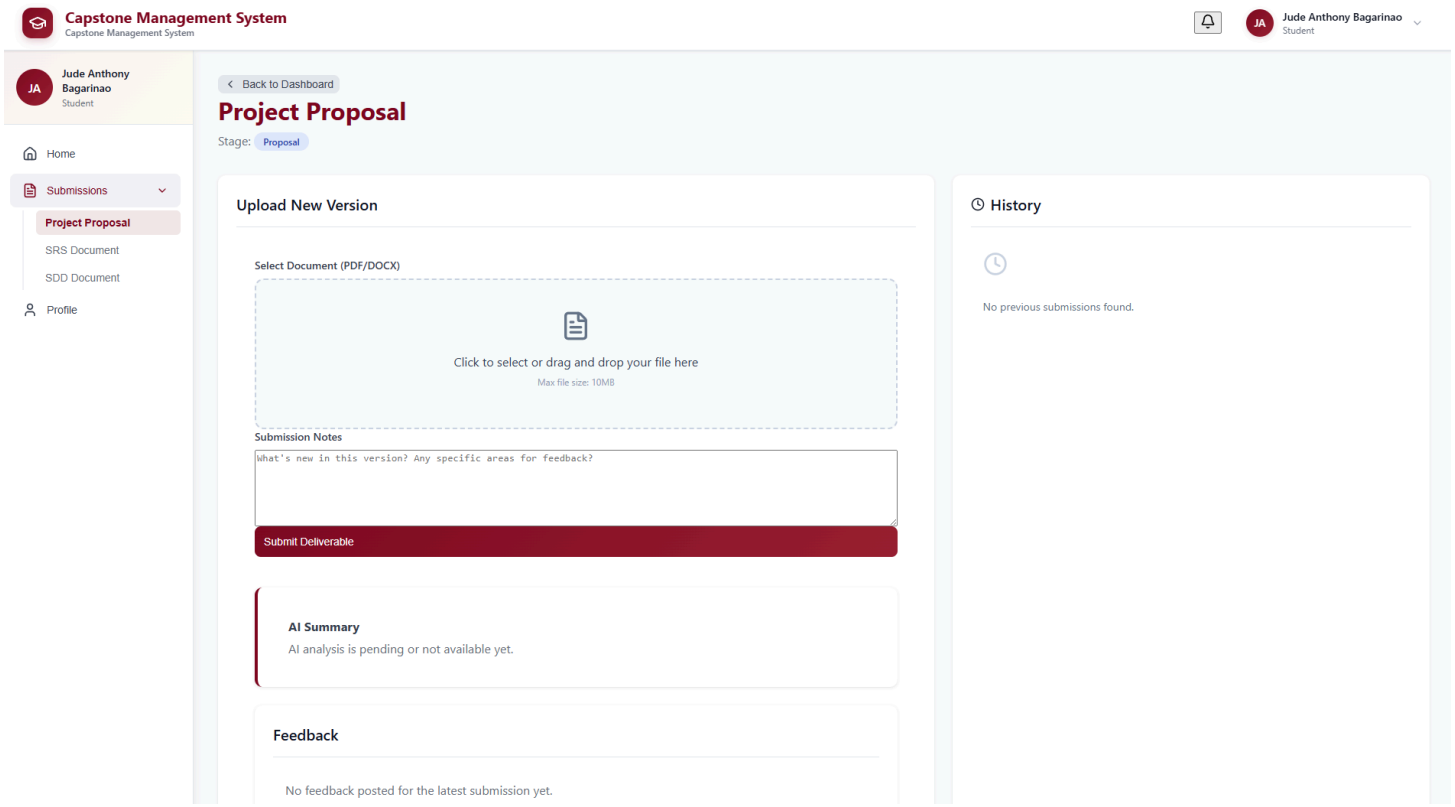


- Data Design
    - ERD or schema


SUBMISSION_PROGRESS
bigint
submission_progress_id
PK
bigint
submission_status_id
FK
double
completion_percentage
double
submission_rate
integer
pending_submissions
integer
late_submissions
datetime
monitored_at
tracks
SUBMISSION_STATUS
bigint
submission_status_id
PK
string
status_type
string
status_description
datetime
submission_timestamp
datetime
deadline_timestamp
integer
revision_count
datetime
created_at
datetime
updated_at
monitors
STATUS_MONITORING_EVENT
bigint
monitoring_event_id
PK
bigint
submission_status_id
FK
string
activity_type
string
monitoring_details
datetime
activity_timestamp
analyzes
DASHBOARD_ANALYTICS
bigint
dashboard_analytics_id
PK
bigint
submission_status_id
FK
double
completion_rate
double
submission_rate
integer
total_submissions
integer
pending_submissions
integer
late_submission_count
datetime
generated_at

2.3 Submission File Management

- User Interface Design



- Front-end component(s)
  - **SubmissionUploadPanel**
    - Description and purpose
      - Provides the user interface for uploading deliverable files and managing submission uploads associated with project milestones and deliverables.
    - Component type or format
      - React Upload Component (components/SubmissionUploadPanel.jsx)
  - **SubmissionVersionPanel**
    - Description and purpose
      - Displays submission revision history and allows users to review different submission versions associated with uploaded deliverables.
    - Component type or format
      - React Version Management Component (components/SubmissionVersionPanel.jsx)
  - **SubmissionCommentSection**
    - Description and purpose

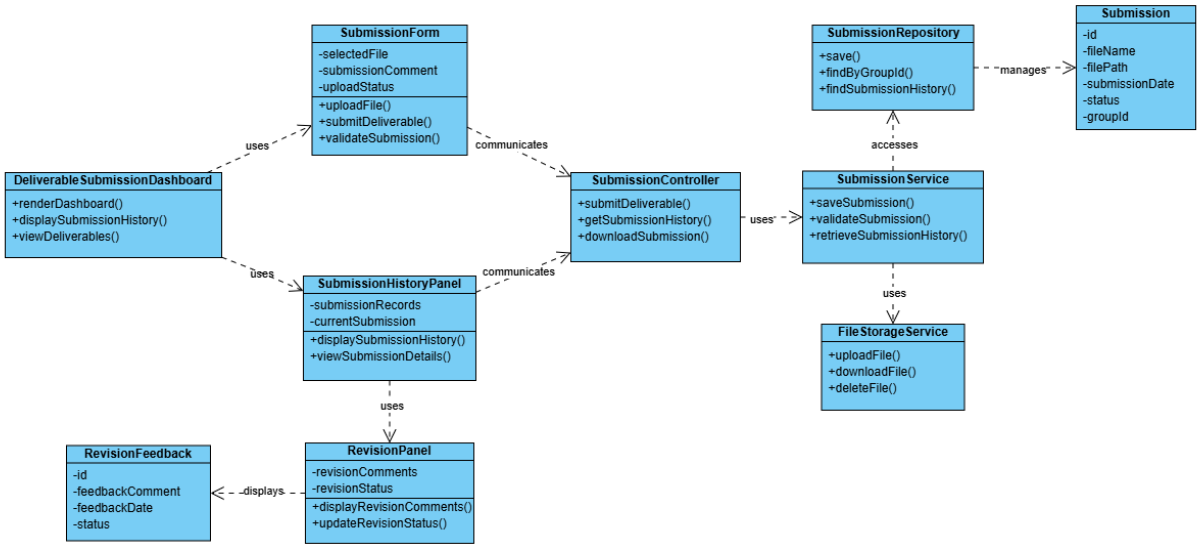
        - Allows users and advisers to view and manage comments associated with deliverable submissions and revision feedback activities.
    - Component type or format
        - React Comment Management Component (components/SubmissionCommentSection.jsx)

- Back-end component(s)
    - **FileStorageService**
        - Description and purpose
            - Handles file upload storage operations for deliverable submissions using local storage or cloud-based storage integration. It also retrieves uploaded submission files when requested.
        - Component type or format
            - Service Class (service/FileStorageService.java)
    - **SubmissionRepository**
        - Description and purpose
            - Retrieves and manages submission records associated with project groups and deliverables including submission timestamps, revision references, and submission status information.
        - Component type or format
            - Repository Interface (repository/SubmissionRepository.java)
    - **SubmissionVersionRepository**
        - Description and purpose
            - Retrieves and manages submission version records used for revision tracking and historical submission management.
        - Component type or format
            - Repository Interface (repository/SubmissionVersionRepository.java)
    - **SubmissionCommentRepository**
        - Description and purpose
            - Retrieves and manages submission comment records associated with adviser feedback and submission discussion activities.
        - Component type or format
            - Repository Interface (repository/SubmissionCommentRepository.java)
    - **Submission**
        - Description and purpose

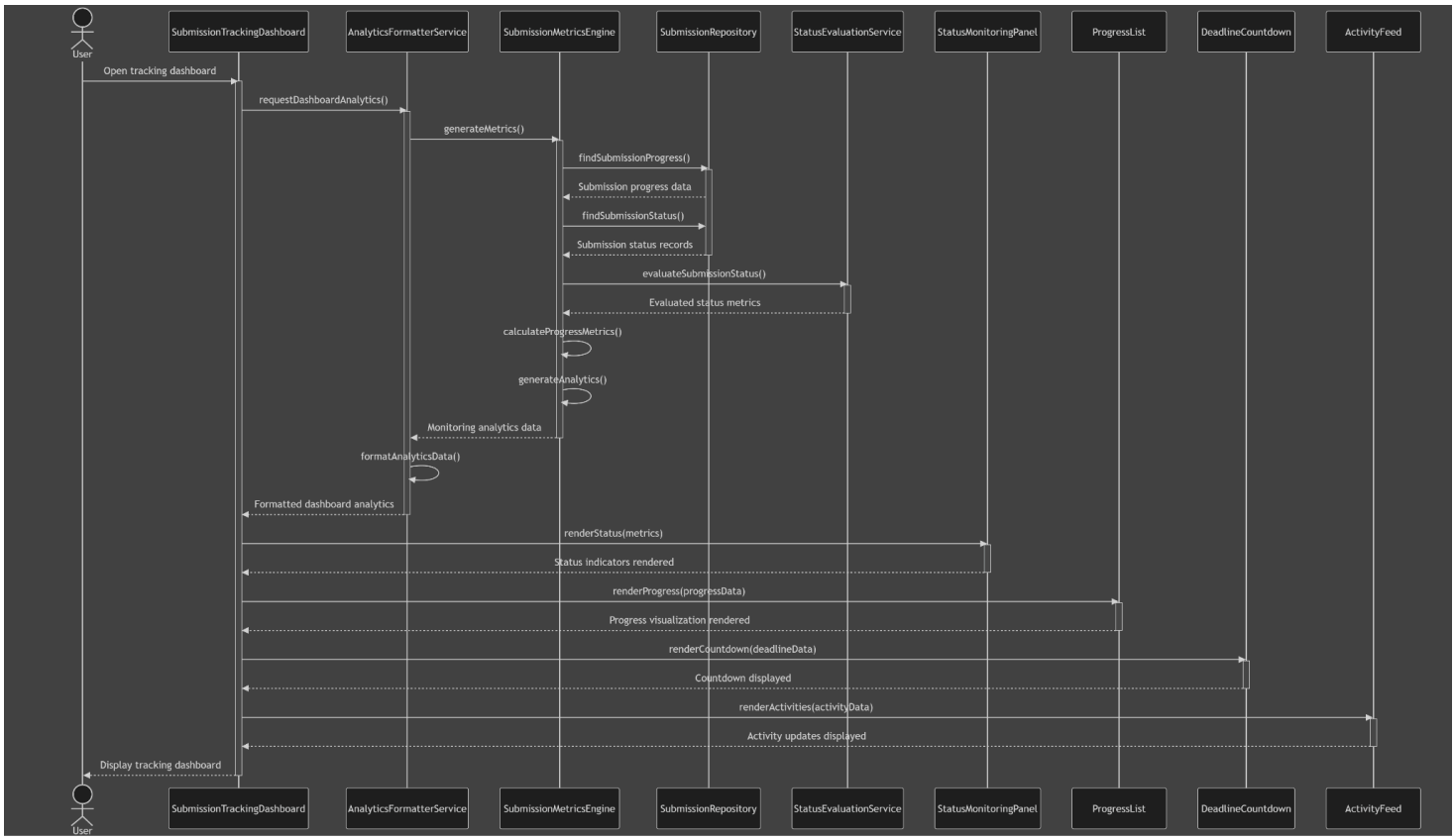
        - Represents submission records associated with project groups and deliverables including submission timestamps, status information, file references, and revision history.
    - Component type or format
        - Entity Class (entity/Submission.java)
- **SubmissionVersion**
    - Description and purpose
        - Represents version history records of submitted deliverables including revision numbers, uploaded file references, and version tracking information.
    - Component type or format
        - Entity Class (entity/SubmissionVersion.java)
- **SubmissionComment**
    - Description and purpose
        - Represents comment records associated with submissions including adviser remarks, feedback discussions, and revision-related communication.
    - Component type or format
        - Entity Class (entity/SubmissionComment.java)
- **AuditService**
    - Description and purpose
        - Records submission-related activities and file management operations for audit logging and system monitoring purposes.
    - Component type or format
        - Service Class (service/AuditService.java)

- Object-Oriented Components
    - Class Diagram

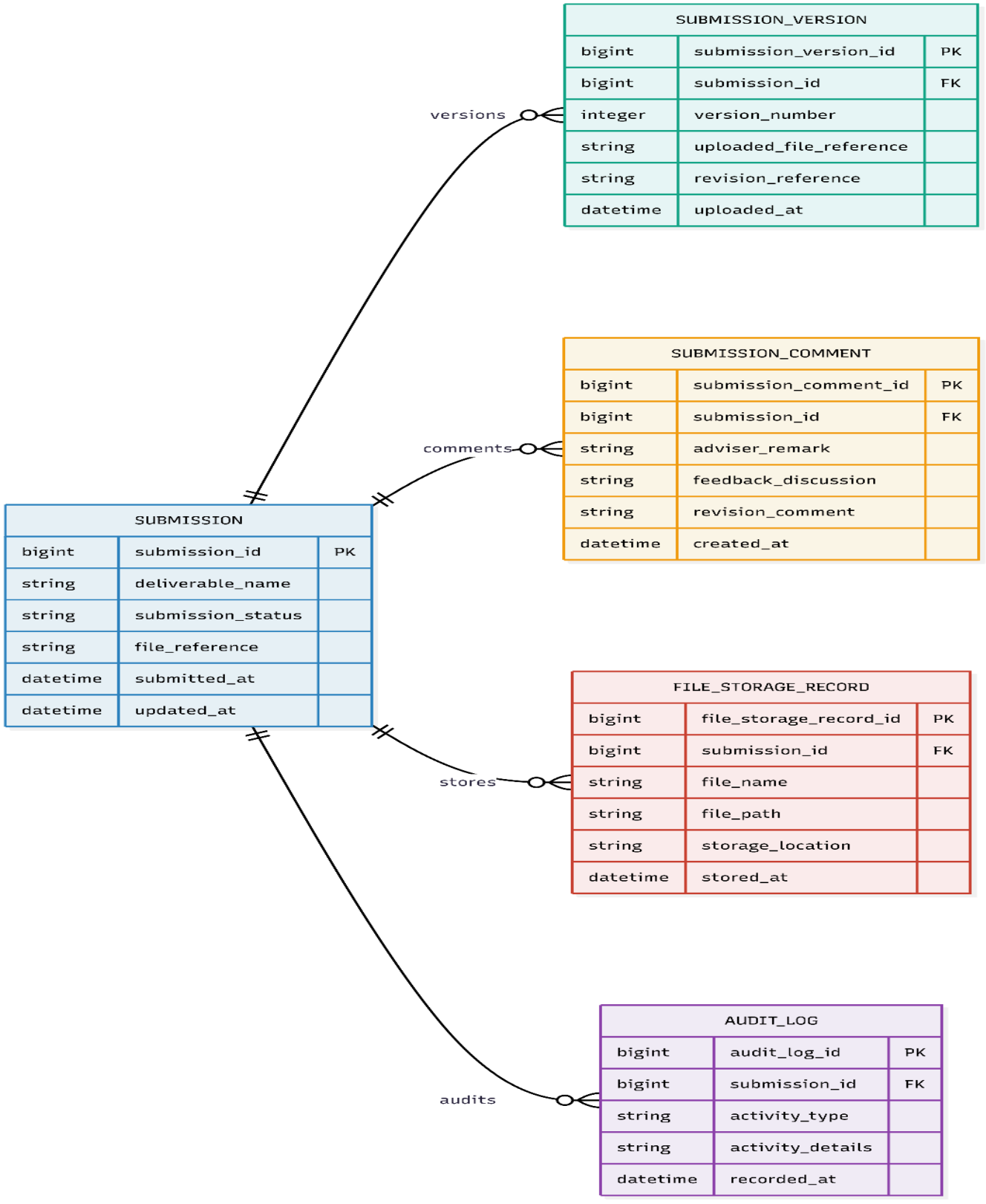
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- Sequence Diagram



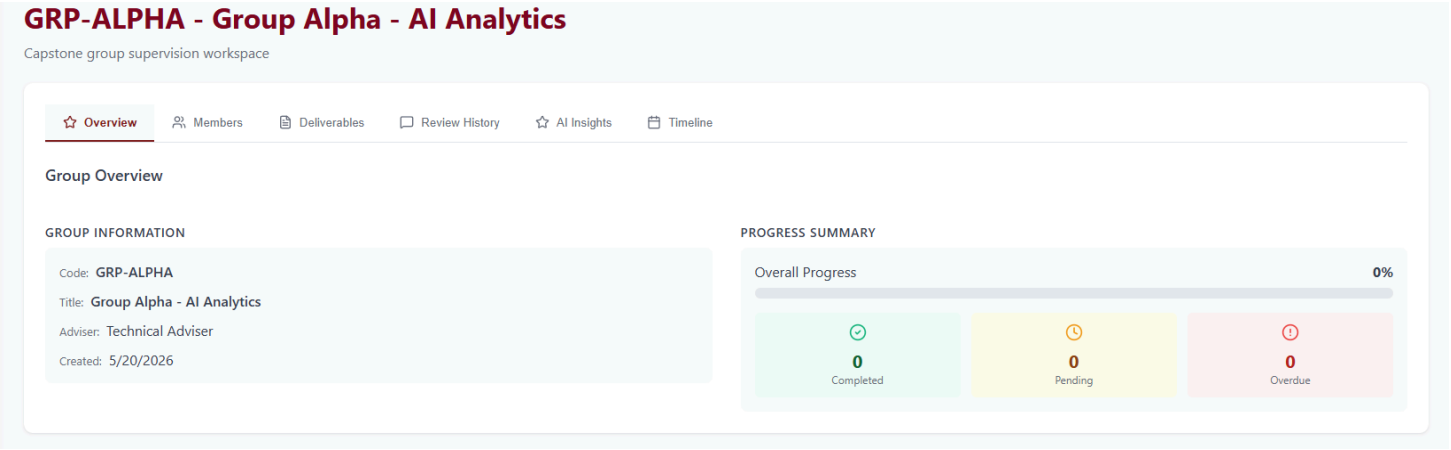
- Data Design
    - ERD or schema


SUBMISSION_VERSION
bigint submission_version_id PK
bigint submission_id FK
integer version_number
string uploaded_file_reference
string revision_reference
datetime uploaded_at
versions
SUBMISSION_COMMENT
bigint submission_comment_id PK
bigint submission_id FK
string adviser_remark
string feedback_discussion
string revision_comment
datetime created_at
comments
SUBMISSION
bigint submission_id PK
string deliverable_name
string submission_status
string file_reference
datetime submitted_at
datetime updated_at
FILE_STORAGE_RECORD
bigint file_storage_record_id PK
bigint submission_id FK
string file_name
string file_path
string storage_location
datetime stored_at
stores
AUDIT_LOG
bigint audit_log_id PK
bigint submission_id FK
string activity_type
string activity_details
datetime recorded_at
audits

Module 3: Analytics & Reporting Dashboard

3.1 AI-Powered Real-Time Submission Insights

- User Interface Design



- Front-end component(s)
  - **InsightHubDashboard**
    - Description and purpose
      - Provides a real-time analytics dashboard for monitoring on-time submissions, pending submissions, late submissions, and milestone completion trends across project groups and deliverables.
    - Component type or format
      - React Analytics Dashboard Component (components/InsightHubDashboard.jsx)
  - **MetricCard**
    - Description and purpose
      - Displays summarized analytics metrics such as on-time percentages, late submission rates, pending submission counts, and milestone performance indicators in a compact dashboard format.
    - Component type or format
      - React Analytics Component (components/MetricCard.jsx)
  - **ProgressList**
    - Description and purpose
      - Displays completion progress information for project groups including submission completion percentages and deliverable tracking summaries used for analytics visualization.
    - Component type or format
      - React Progress Visualization Component

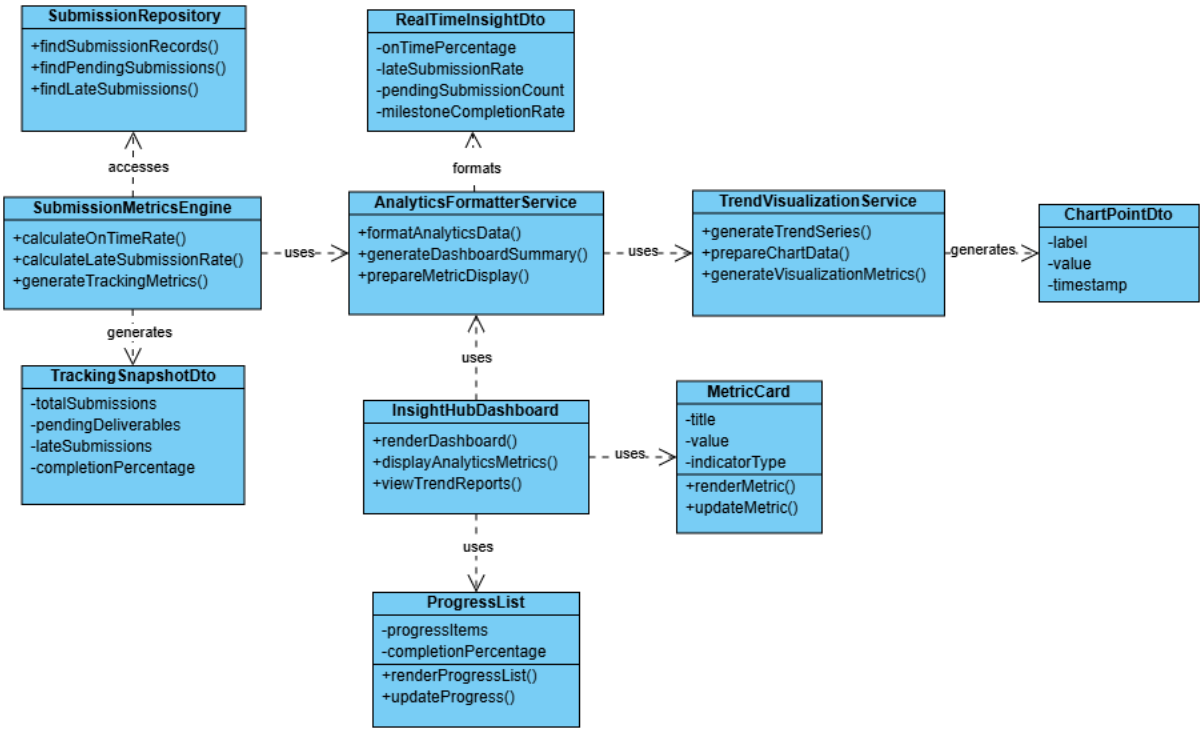
(components/ProgressList.jsx)

- Back-end component(s)
  - **SubmissionMetricsEngine**
    - Description and purpose
      - Processes submission records and computes real-time analytics metrics including on-time completion rates, pending submissions, late submission percentages, and milestone tracking statistics.
    - Component type or format
      - Service Class (service/SubmissionMetricsEngine.java)
  - **AnalyticsFormatterService**
    - Description and purpose
      - Formats processed analytics data into structured dashboard representations used by frontend analytics and monitoring components.
    - Component type or format
      - Service Class (service/AnalyticsFormatterService.java)
  - **TrendVisualizationService**
    - Description and purpose
      - Transforms computed analytics information into trend visualization data structures used for chart rendering and dashboard reporting interfaces.
    - Component type or format
      - Service Class (service/TrendVisualizationService.java)
  - **SubmissionRepository**
    - Description and purpose
      - Retrieves submission records and monitoring data required for generating real-time submission analytics and milestone tracking insights.
    - Component type or format
      - Repository Interface (repository/SubmissionRepository.java)
  - **RealTimeInsightDto**
    - Description and purpose
      - Represents structured real-time analytics information including submission performance metrics, trend series data, and status breakdown summaries used for dashboard reporting.
    - Component type or format
      - Data Transfer Object (dto/RealTimeInsightDto.java)
  - **ChartPointDto**

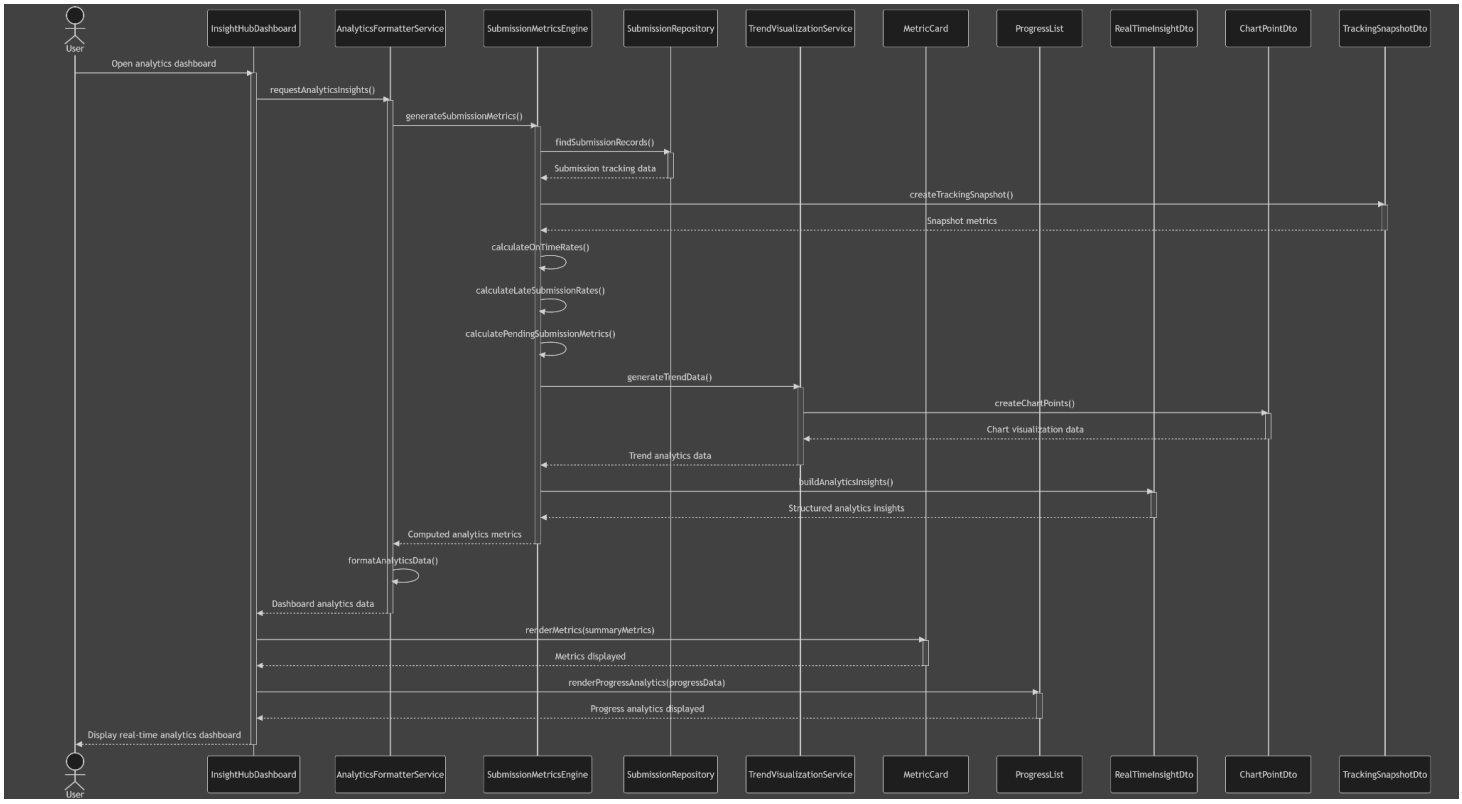
        - Description and purpose
            - Represents chart visualization data points used for displaying analytics trends, milestone completion rates, and submission monitoring statistics within graphical dashboard components.
        - Component type or format
            - Data Transfer Object (dto/ChartPointDto.java)
    - **TrackingSnapshotDto**
        - Description and purpose
            - Represents summarized submission tracking information including submission counts, pending deliverables, late submissions, and project group progress metrics used for analytics computation.
        - Component type or format
            - Data Transfer Object (dto/TrackingSnapshotDto.java)

- Object-Oriented Components
    - Class Diagram

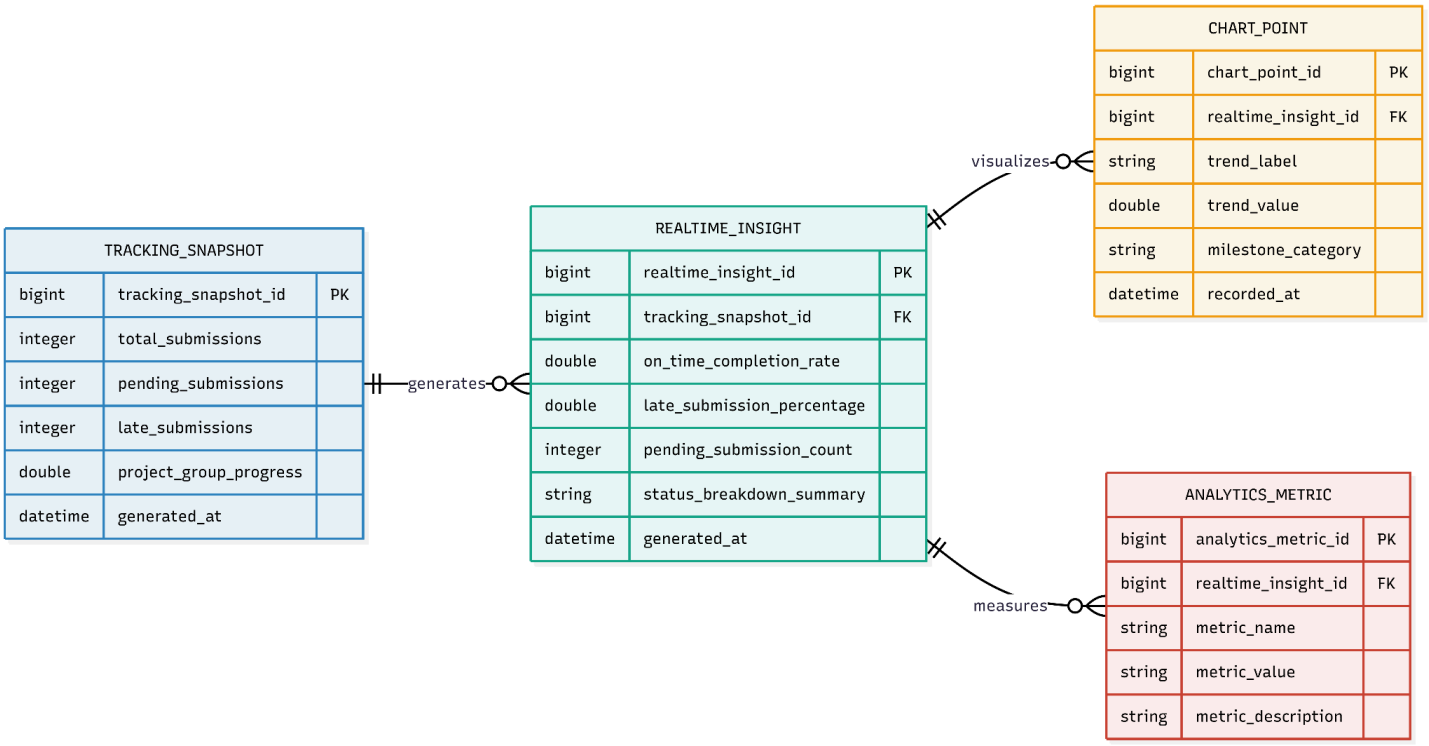
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- Sequence Diagram

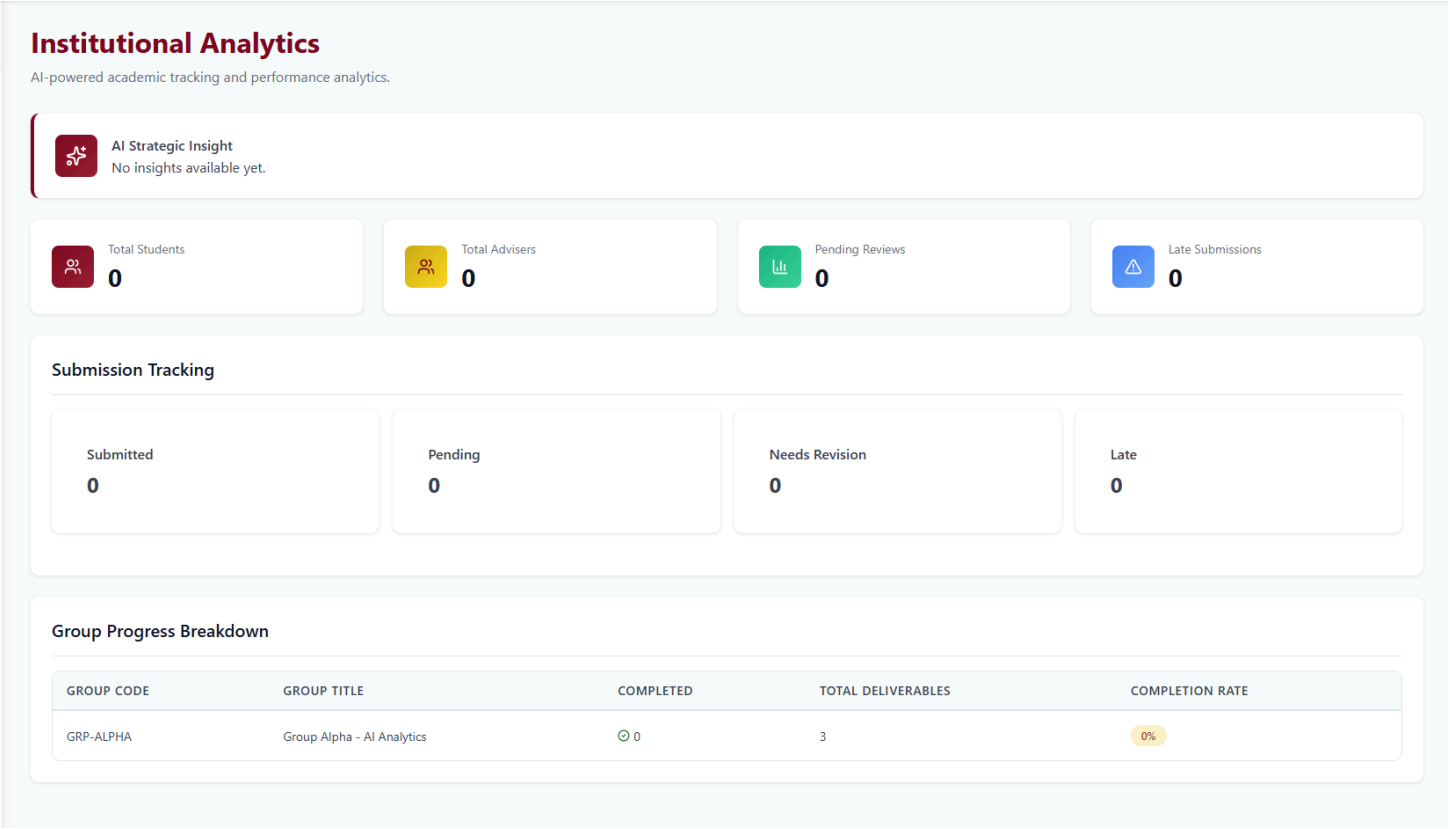


- Data Design
    - ERD or schema



3.2 AI Submission Summary

- User Interface Design



- Front-end component(s)
  - **InstitutionalOversightDashboard**
    - Description and purpose
      - Provides an institutional-level dashboard for reviewing AI-generated submission summaries, deliverable progress, revision history, submission timelines, and overall project group performance information.
    - Component type or format
      - React Analytics Dashboard Component (components/InstitutionalOversightDashboard.jsx)
  - **ActivityFeed**
    - Description and purpose
      - Displays chronological submission-related activities including submission events, revision updates, feedback records, and timeline monitoring information associated with project deliverables.
    - Component type or format
      - React Activity Visualization Component (components/ActivityFeed.jsx)

- Back-end component(s)
  - **SummaryFormatterService**
    - Description and purpose
      - Processes and formats submission-related information into structured AI-generated summary reports and institutional overview data used for dashboard presentation.
    - Component type or format
      - Service Class (service/SummaryFormatterService.java)
  - **SubmissionRepository**
    - Description and purpose
      - Retrieves submission records, submission status information, and deliverable tracking data required for generating AI-assisted submission summaries and institutional reporting insights.
    - Component type or format
      - Repository Interface (repository/SubmissionRepository.java)
  - **SubmissionVersionRepository**
    - Description and purpose
      - Retrieves submission revision history and version tracking records used for generating submission timeline summaries and revision monitoring analytics.
    - Component type or format
      - Repository Interface (repository/SubmissionVersionRepository.java)
  - **SubmissionSummaryDto**
    - Description and purpose
      - Represents formatted submission summary information including project group details, deliverable summaries, revision history, and timeline monitoring data used for frontend dashboard presentation.
    - Component type or format
      - Data Transfer Object (dto/SubmissionSummaryDto.java)
  - **RawSubmissionSummaryData**
    - Description and purpose

- Represents raw submission-related information collected from deliverables, revision records, and timeline data before summary formatting and AI-assisted reporting generation.

▪ Component type or format

- Data Transfer Object (dto/RawSubmissionSummaryData.java)

o SummaryInsightDto

▪ Description and purpose

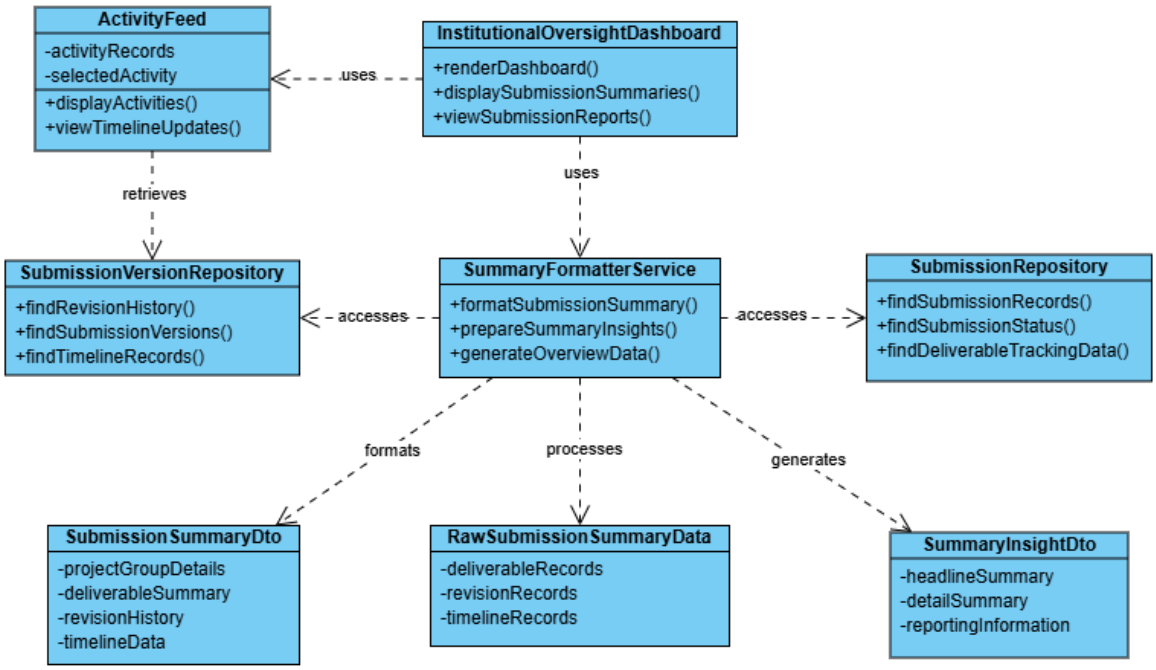
- Represents structured AI-generated summary insights including headline summaries, detail summaries, and institutional reporting information related to submission progress and deliverable monitoring.

▪ Component type or format

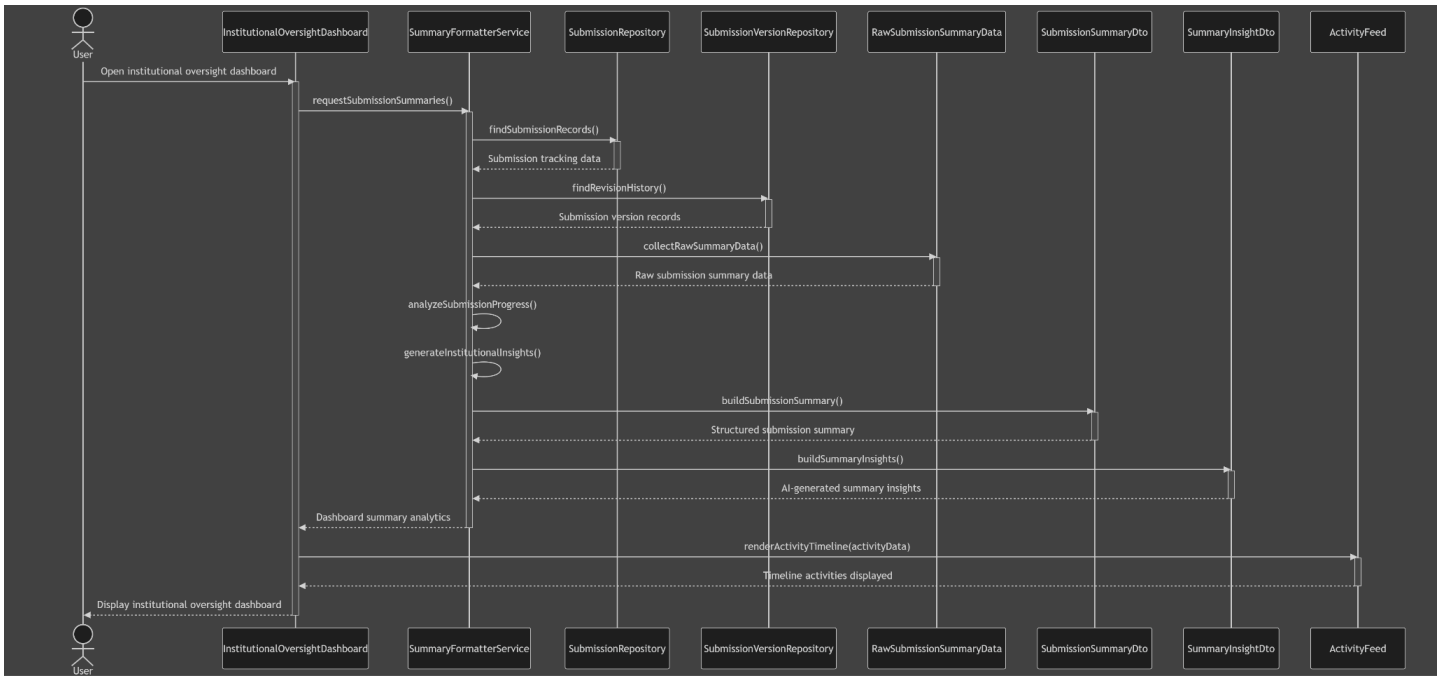
- Data Transfer Object (dto/SummaryInsightDto.java)

- Object-Oriented Components
  - o Class Diagram

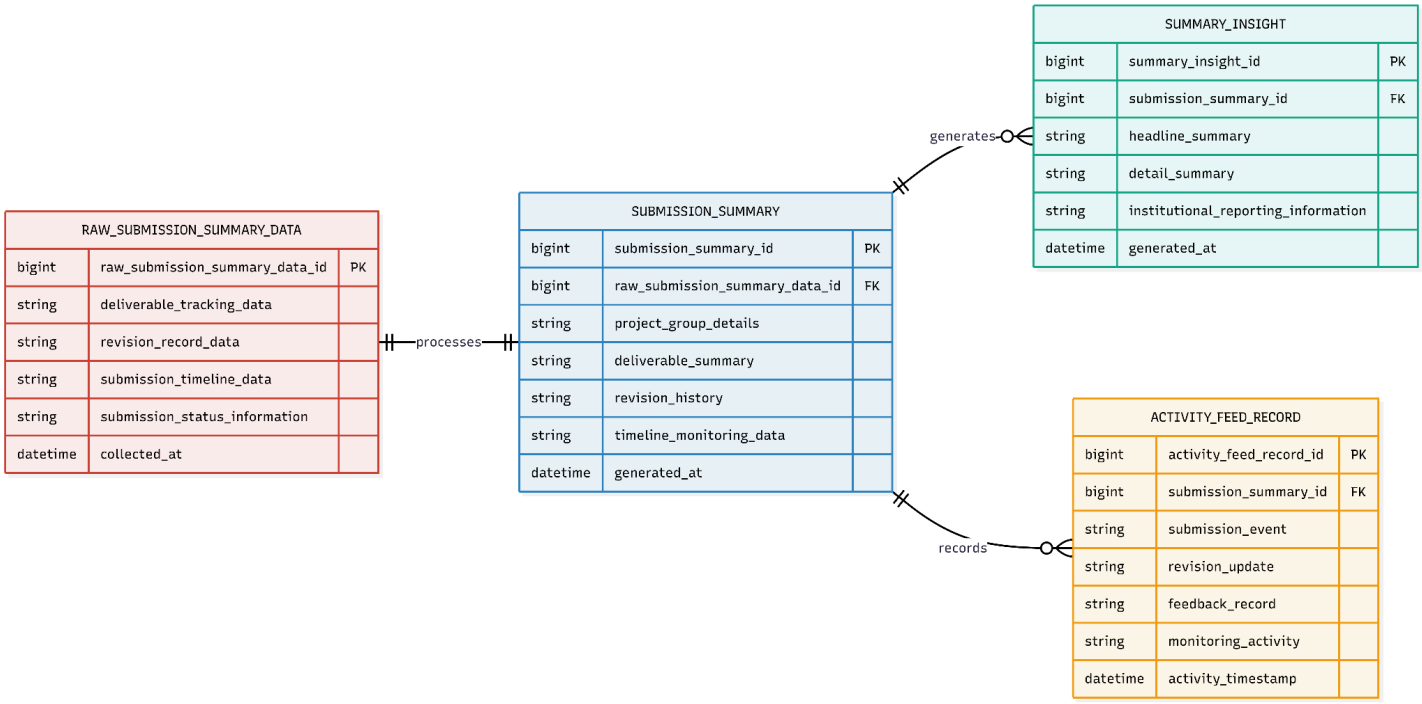
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- Sequence Diagram

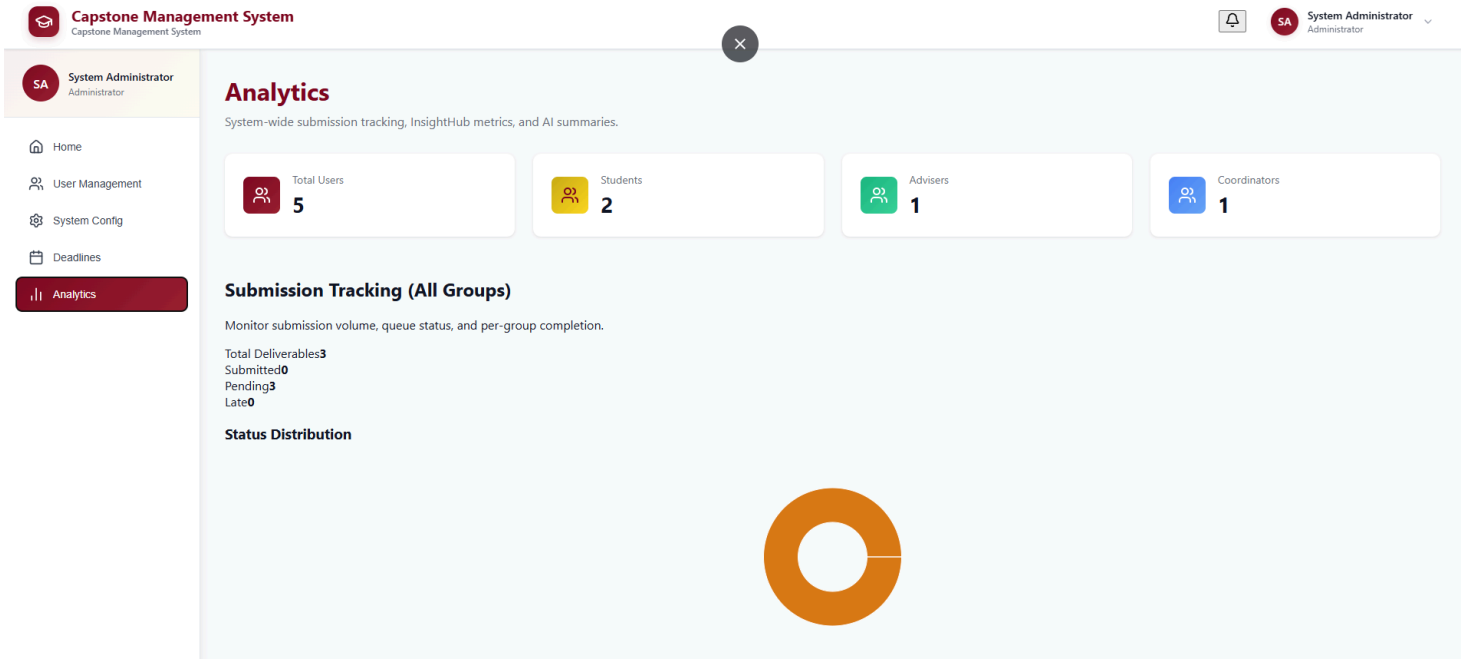


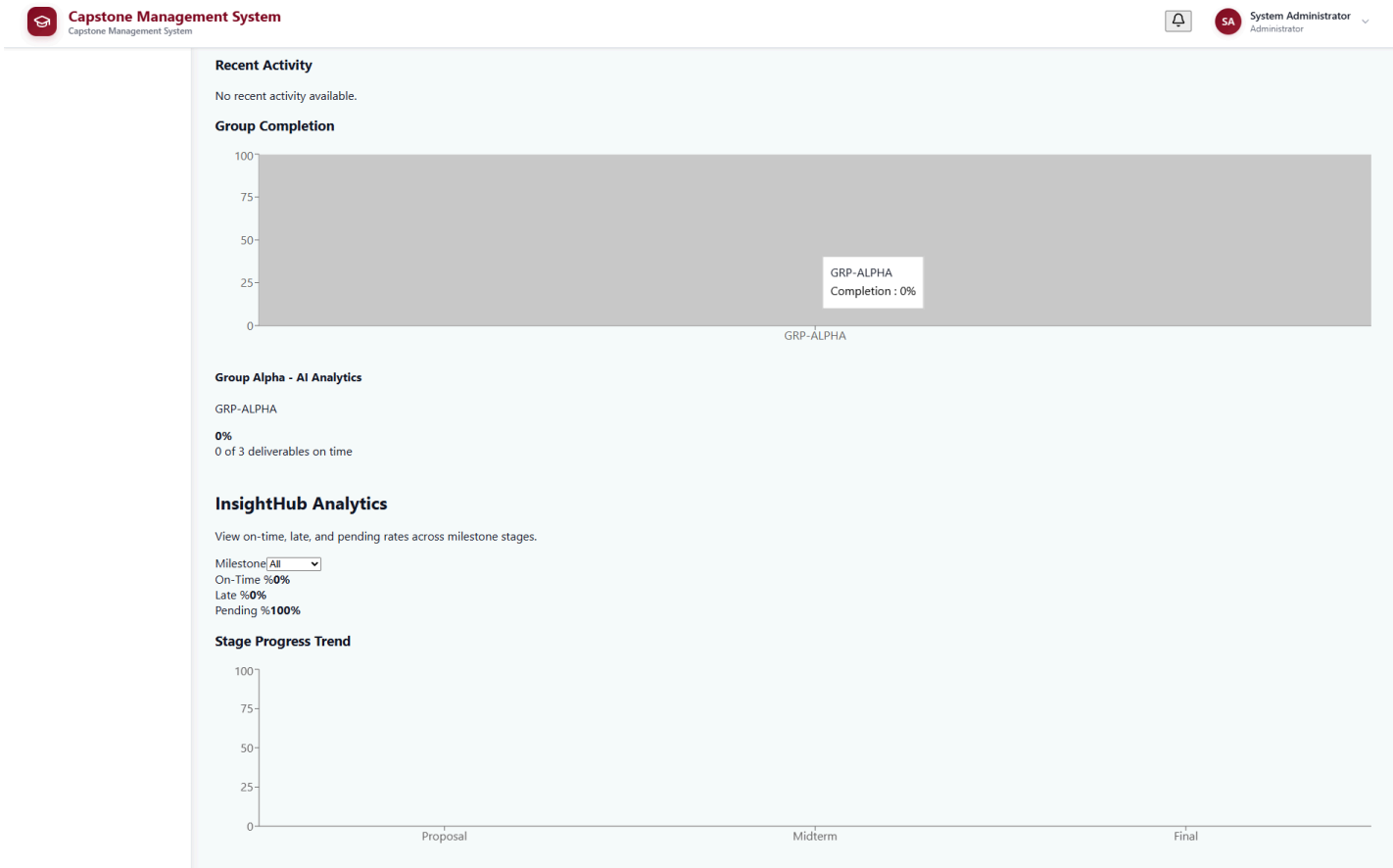
- Data Design
    - ERD or schema

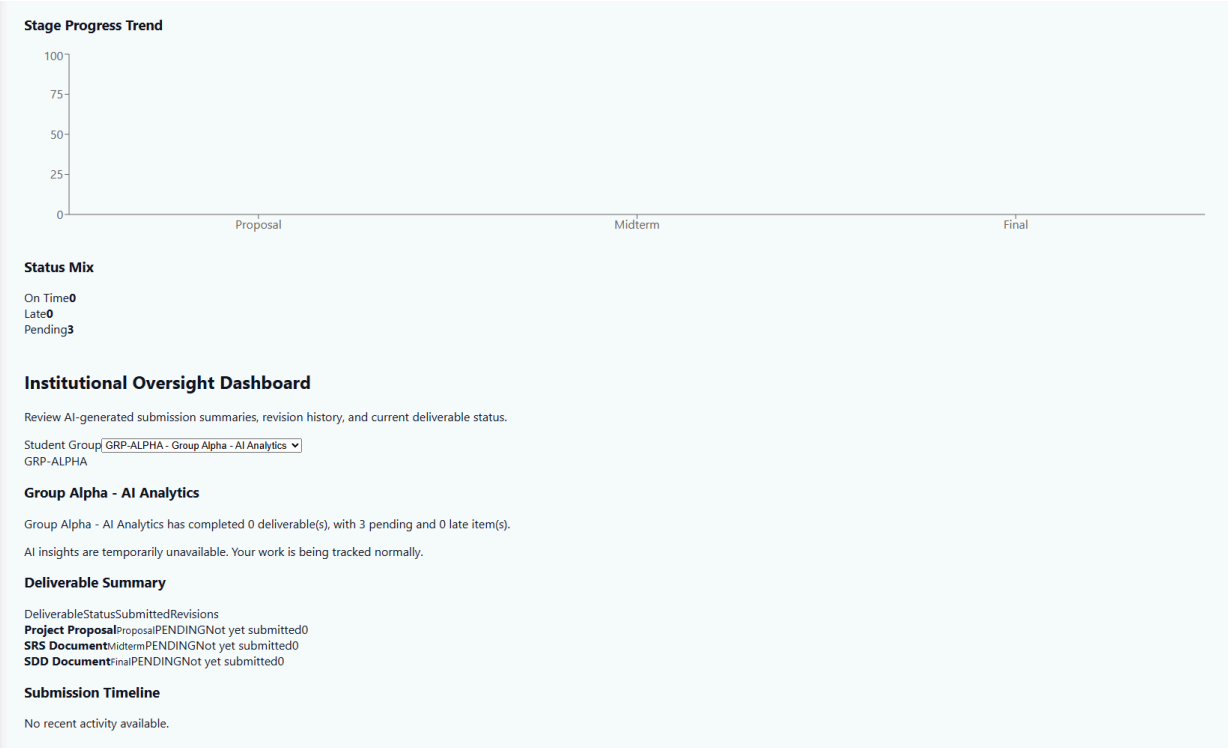


3.3 Submission Tracking Analytics Dashboard

- User Interface Design


Capstone Management System
Capstone Management System
System Administrator
Administrator
SA
System Administrator
Administrator
Home
User Management
System Config
Deadlines
Analytics
Analytics
System-wide submission tracking, InsightHub metrics, and AI summaries.
Total Users
5
Students
2
Advisers
1
Coordinators
1
Submission Tracking (All Groups)
Monitor submission volume, queue status, and per-group completion.
Total Deliverables3
Submitted0
Pending3
Late0
Status Distribution


Capstone Management System
Capstone Management System
System Administrator
Administrator
Recent Activity
No recent activity available.
Group Completion
100
75
50
25
0
GRP-ALPHA
Completion : 0%
GRP-ALPHA
Group Alpha - AI Analytics
GRP-ALPHA
0%
0 of 3 deliverables on time
InsightHub Analytics
View on-time, late, and pending rates across milestone stages.
Milestone All
On-Time %0%
Late %0%
Pending %100%
Stage Progress Trend
100
75
50
25
0
Proposal
Midterm
Final



- Front-end component(s)
  - **SubmissionTrackingDashboard**
    - Description and purpose
      - Provides a dashboard interface for monitoring submission analytics including submission distribution, deliverable completion rates, recent submission activities, and project group progress tracking.
    - Component type or format
      - React Analytics Dashboard Component (components/SubmissionTrackingDashboard.jsx)
  - **MetricCard**
    - Description and purpose
      - Displays summarized analytics metrics such as total deliverables, submitted deliverables, pending submissions, and late submission counts in a compact dashboard visualization format.
    - Component type or format
      - React Analytics Component (components/MetricCard.jsx)
  - **ProgressList**
    - Description and purpose

- Displays completion progress information for project groups including completion percentages, submitted deliverable counts, and overall submission tracking summaries.

- Component type or format
  - React Progress Visualization Component (components/ProgressList.jsx)

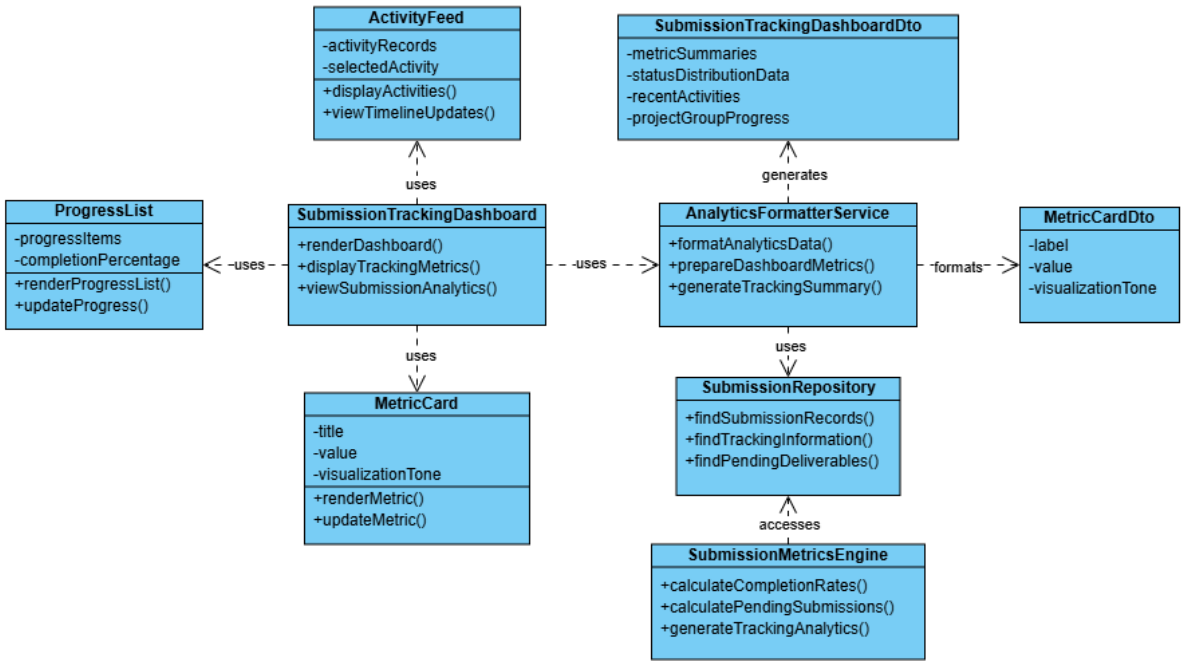
- **ActivityFeed**
  - Description and purpose
    - Displays recent submission-related activities, timeline events, and monitoring updates associated with deliverable tracking and submission analytics.
  - Component type or format
    - React Activity Visualization Component (components/ActivityFeed.jsx)

- Back-end component(s)
  - **SubmissionMetricsEngine**
    - Description and purpose
      - Processes submission records and computes analytics metrics related to submission tracking, completion rates, pending deliverables, late submissions, and project group monitoring information.
    - Component type or format
      - Service Class (service/SubmissionMetricsEngine.java)
  - **AnalyticsFormatterService**
    - Description and purpose
      - Formats processed submission analytics data into structured dashboard representations used for frontend tracking visualization and reporting components.
    - Component type or format
      - Service Class (service/AnalyticsFormatterService.java)
  - **SubmissionRepository**
    - Description and purpose
      - Retrieves submission records and monitoring information required for submission tracking analytics, progress computation, and dashboard reporting activities.
    - Component type or format
      - Repository Interface (repository/SubmissionRepository.java)

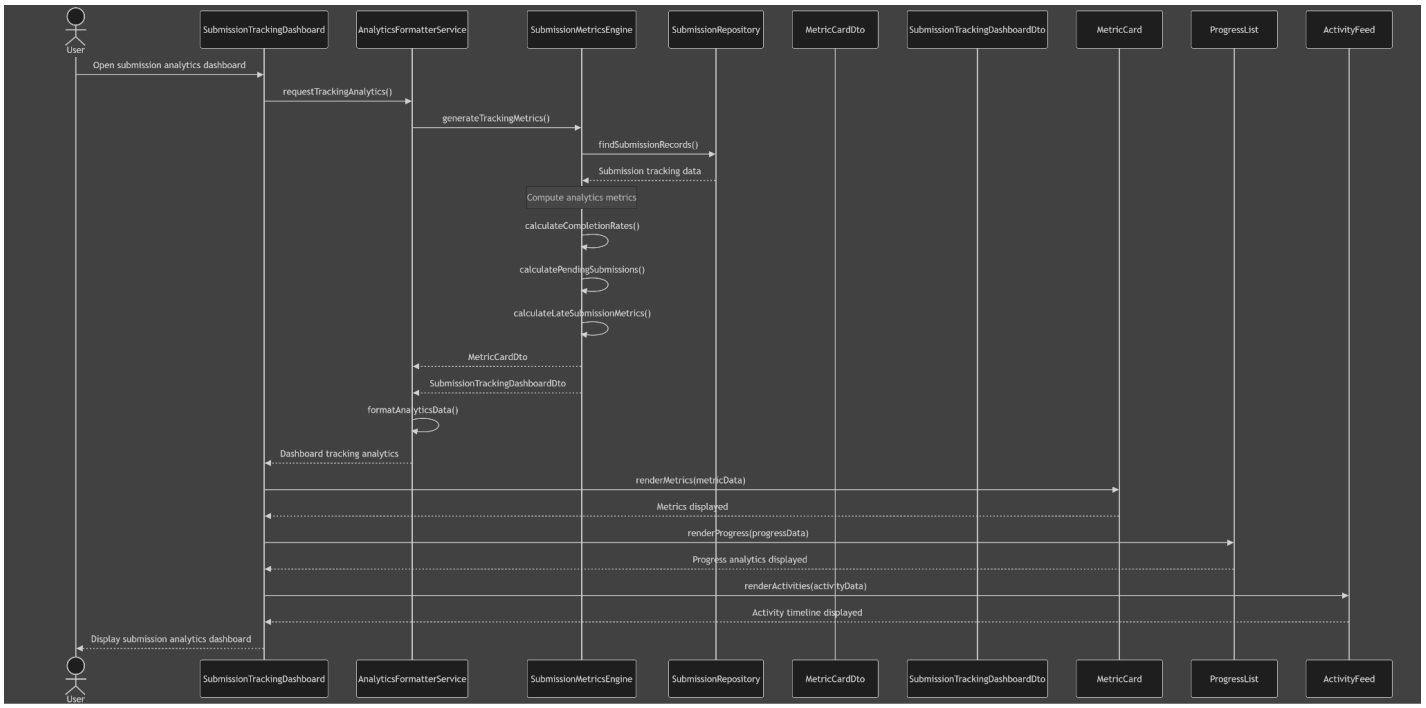
- **MetricCardDto**
  - Description and purpose
    - Represents summarized analytics metric information displayed within dashboard metric cards including labels, values, and visualization tones.
  - Component type or format
    - Data Transfer Object (dto/MetricCardDto.java)
- **SubmissionTrackingDashboardDto**
  - Description and purpose
    - Represents structured submission tracking dashboard information including metric summaries, status distribution data, recent activity feeds, and project group progress analytics.
  - Component type or format
    - Data Transfer Object (dto/SubmissionTrackingDashboardDto.java)

- Object-Oriented Components
  - Class Diagram

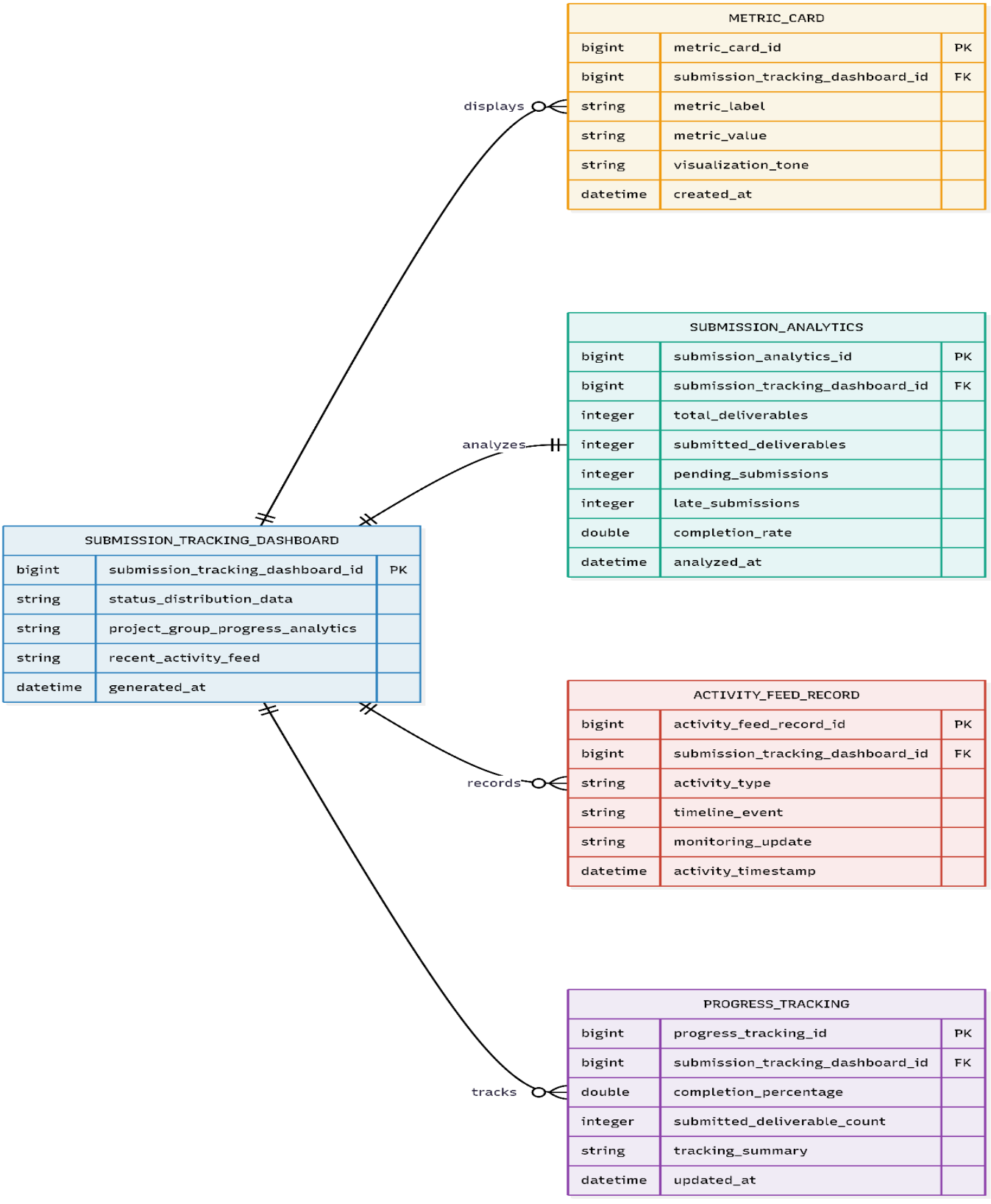
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- Sequence Diagram

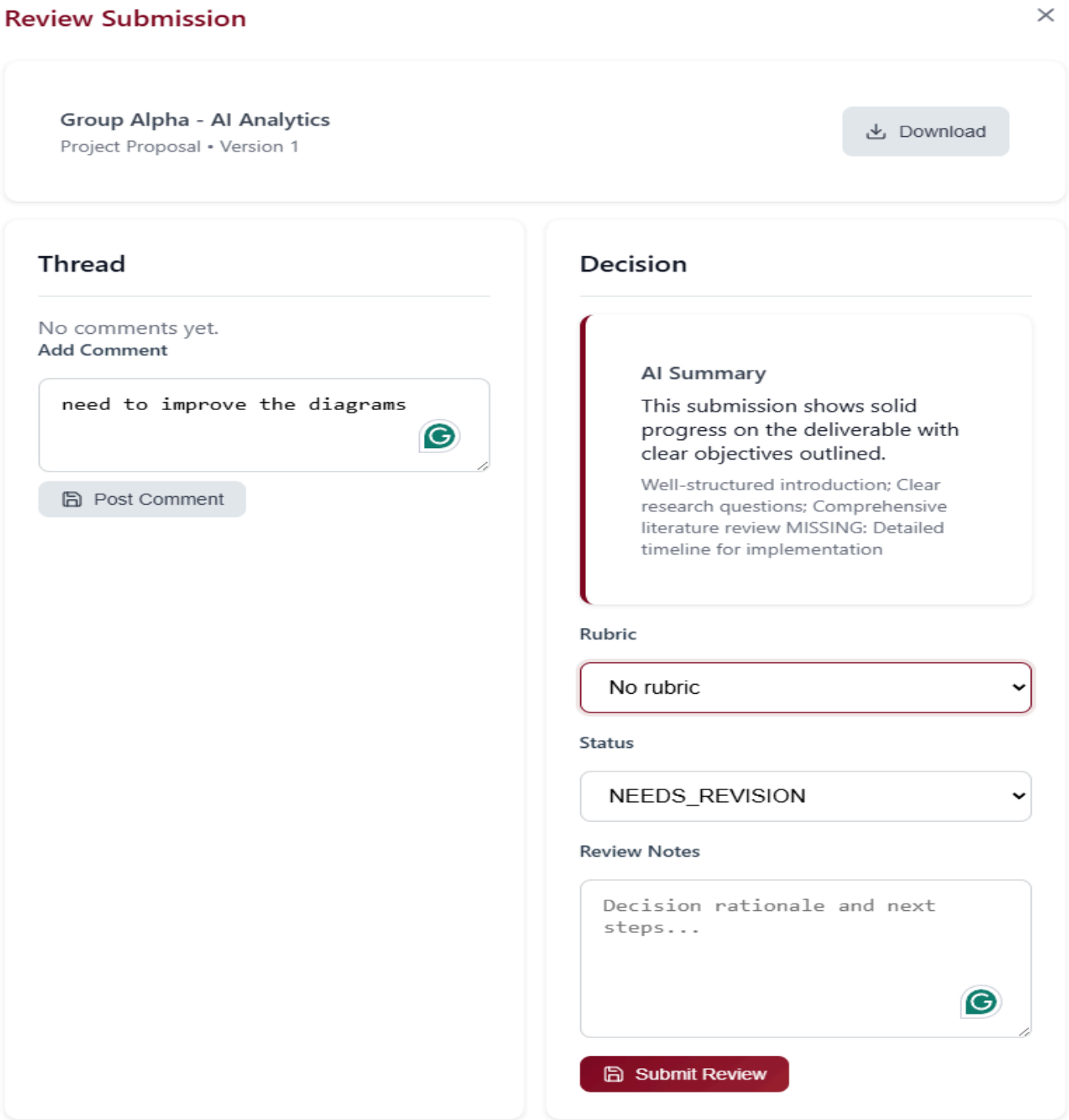


- Data Design
    - ERD or schema


METRIC_CARD
bigint metric_card_id PK
bigint submission_tracking_dashboard_id FK
string metric_label
string metric_value
string visualization_tone
datetime created_at
displays
SUBMISSION_ANALYTICS
bigint submission_analytics_id PK
bigint submission_tracking_dashboard_id FK
integer total_deliverables
integer submitted_deliverables
integer pending_submissions
integer late_submissions
double completion_rate
datetime analyzed_at
analyzes
SUBMISSION_TRACKING_DASHBOARD
bigint submission_tracking_dashboard_id PK
string status_distribution_data
string project_group_progress_analytics
string recent_activity_feed
datetime generated_at
ACTIVITY_FEED_RECORD
bigint activity_feed_record_id PK
bigint submission_tracking_dashboard_id FK
string activity_type
string timeline_event
string monitoring_update
datetime activity_timestamp
records
PROGRESS_TRACKING
bigint progress_tracking_id PK
bigint submission_tracking_dashboard_id FK
double completion_percentage
integer submitted_deliverable_count
string tracking_summary
datetime updated_at
tracks

3.4 Evaluation and Rubric Feedback System

- User Interface Design



- Front-end component(s)
    - **AdviserDocumentReview**
        - Description and purpose
            - Provides advisers with an interface for reviewing submitted deliverables, evaluating project submissions, and managing approval or rejection decisions for student groups.
        - Component type or format
            - React Review Management Component (pages/adviser/AdviserDocumentReview.jsx)
- Back-end component(s)
    - **FeedbackService**
        - Description and purpose
            - Handles adviser evaluation logic including submission review processing, rubric evaluation management, feedback storage, score computation, and submission status updates.
        - Component type or format
            - Service Class (service/FeedbackService.java)
    - **AdviserFeedbackRepository**
        - Description and purpose
            - Retrieves and manages adviser evaluation records, feedback information, rubric assessments, and criterion evaluation results associated with submissions.
        - Component type or format
            - Repository Interface (repository/AdviserFeedbackRepository.java)
    - **RubricRepository**
        - Description and purpose
            - Retrieves and manages rubric records used for adviser evaluation, grading criteria definition, and submission assessment activities.
        - Component type or format
            - Repository Interface (repository/RubricRepository.java)
    - **RubricCriterionRepository**
        - Description and purpose
            - Retrieves and manages rubric criterion records used for detailed scoring

and criterion-based adviser evaluation processes.

- Component type or format
    - Repository Interface (repository/RubricCriterionRepository.java)

- **AdviserFeedback**
    - Description and purpose
        - Represents adviser evaluation records including comments, total scores, evaluation timestamps, associated submissions, and rubric-based assessment information.
    - Component type or format
        - Entity Class (entity/AdviserFeedback.java)

- **CriterionEvaluation**
    - Description and purpose
        - Represents criterion-based evaluation records including assigned scores, rubric criteria references, and adviser assessment results associated with submission evaluations.
    - Component type or format
        - Entity Class (entity/CriterionEvaluation.java)

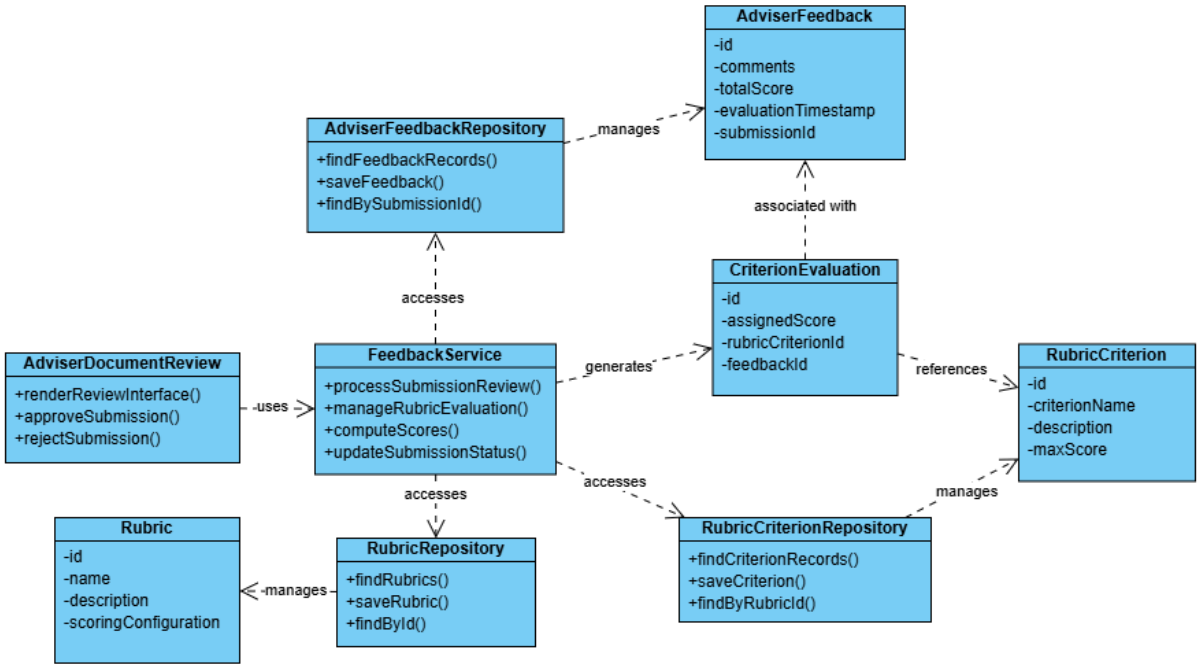
- **Rubric**
    - Description and purpose
        - Represents rubric structures used for adviser grading and submission evaluation including rubric names, descriptions, scoring configurations, and associated criteria.
    - Component type or format
        - Entity Class (entity/Rubric.java)

- **RubricCriterion**
    - Description and purpose
        - Represents individual rubric criteria used for detailed submission assessment and criterion-based grading during adviser evaluations.

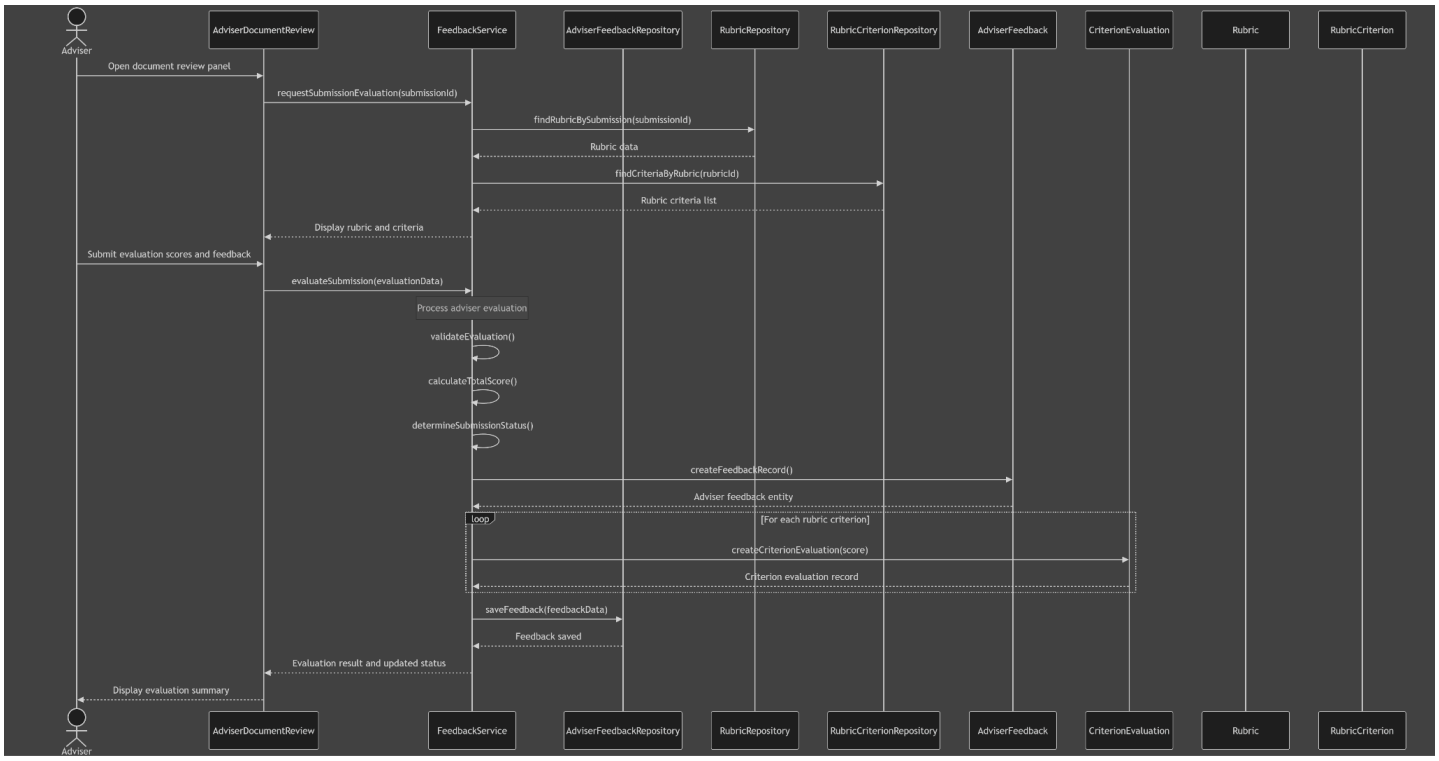
- Component type or format
    - Entity Class (entity/RubricCriterion.java)

- Object-Oriented Components
    - Class Diagram

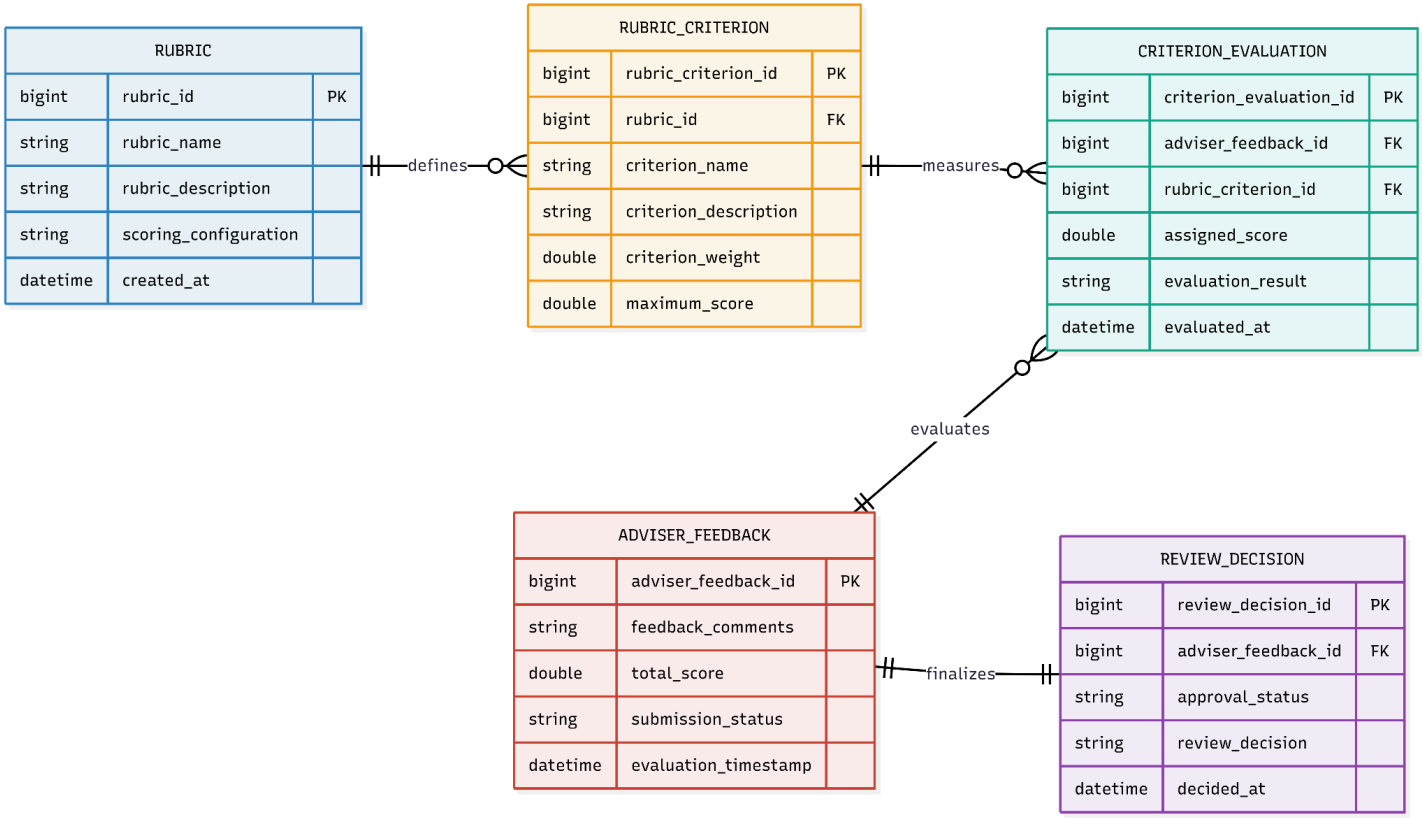
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    - Sequence Diagram



- Data Design
    - ERD or schema


RUBRIC
bigint rubric_id PK
string rubric_name
string rubric_description
string scoring_configuration
datetime created_at
defines
RUBRIC_CRITERION
bigint rubric_criterion_id PK
bigint rubric_id FK
string criterion_name
string criterion_description
double criterion_weight
double maximum_score
measures
CRITERION_EVALUATION
bigint criterion_evaluation_id PK
bigint adviser_feedback_id FK
bigint rubric_criterion_id FK
double assigned_score
string evaluation_result
datetime evaluated_at
evaluates
ADVISER_FEEDBACK
bigint adviser_feedback_id PK
string feedback_comments
double total_score
string submission_status
datetime evaluation_timestamp
finalizes
REVIEW_DECISION
bigint review_decision_id PK
bigint adviser_feedback_id FK
string approval_status
string review_decision
datetime decided_at